Fourth Year Pharmacy Students – Assiut University
Elective Medicinal Chemistry Course – First Semester

Drug Design and Discovery
Lecture 1

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DRUG DESIGN AND DEVELOPMENT

Stages

1) Identify target disease
2) Identify drug target
3) Establish testing procedures
4) Find a lead compound
5) Structure Activity Relationships (SAR)
6) Identify a pharmacophore
7) Drug design- optimising target interactions
8) Drug design - optimising pharmacokinetic properties
9) Toxicological and safety tests
10) Chemical development and production
11) Patenting and regulatory affairs
12) Clinical trials

Why drug design?

The drug should have a good selectivity for its target

2. The drug should have a good level of activity for its target

3. The drug should have minimum side effects

4. The drug should be easily synthesized

5. The drug should be chemically stable

6. The drug should have acceptable pharmacokinetics properties

7. The drug should be non-toxic

Aspects in drug design

1.Pharmacodynamics properties:
To improve the drug's ability to reach its target

2.Pharmacokinetics properties:
To have acceptable life time .

Improving Pharmacodynamics properties:

To optimise binding interactions with target and induce selectivity

Strategies:

1-Vary alkyl substituents.

2-Vary aryl substituents.

3-Extension.

4-Chain extensions / contractions.

5-Ring expansions / contractions.

6-Ring variation.

7-Isosteres.

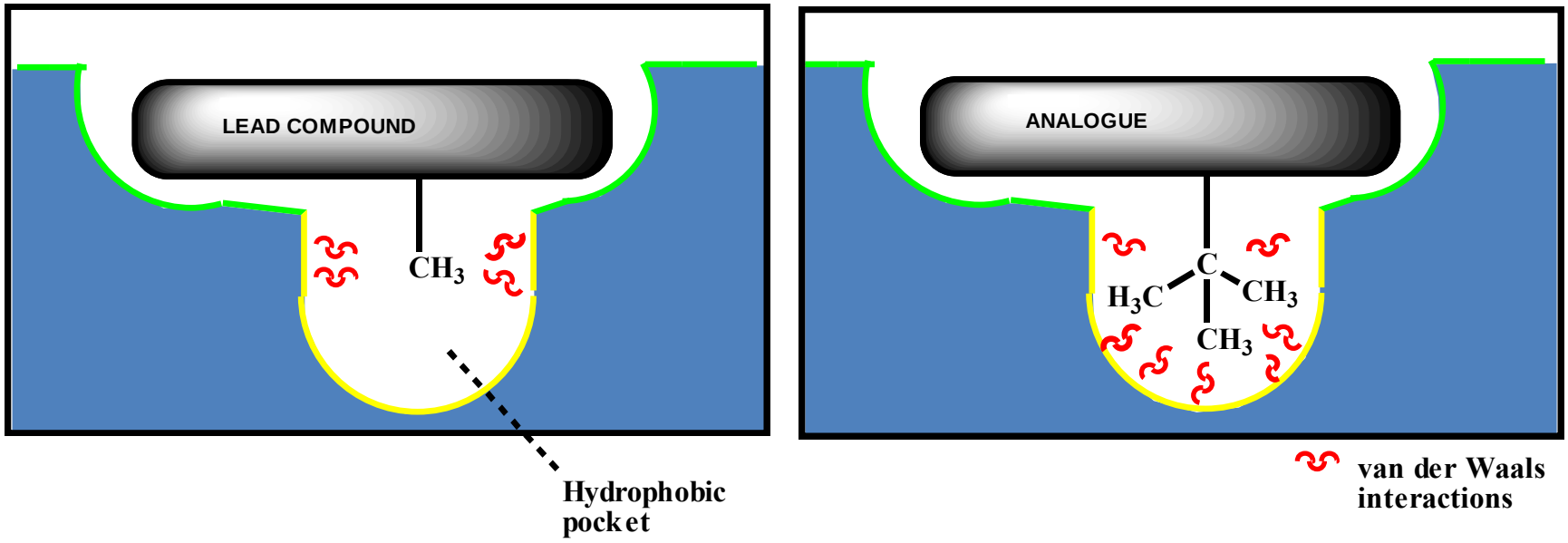
8-Simplification.

9-Rigidification.

1 - Vary Alkyl Substituents

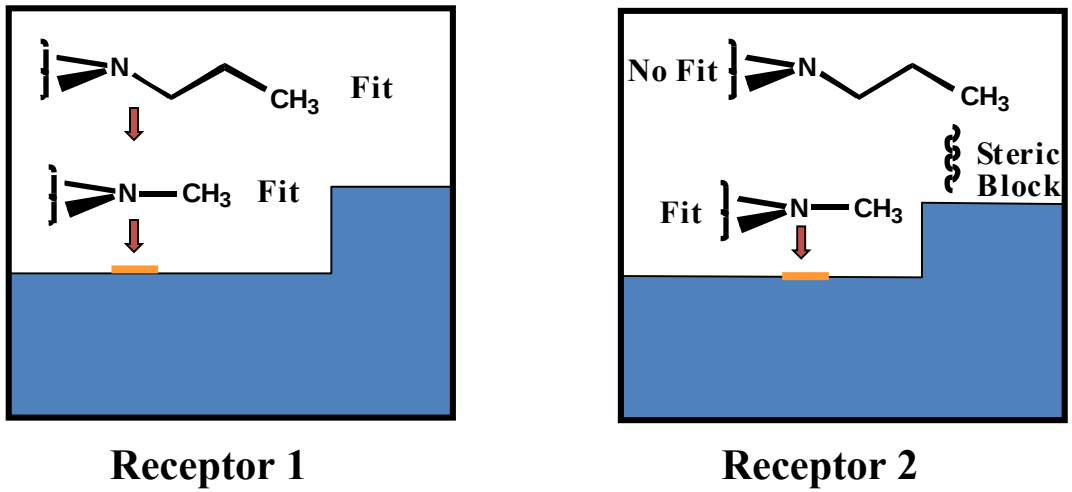
1-To optimise interaction

Alkyl group in lead compound may interact with hydrophobic region in binding site

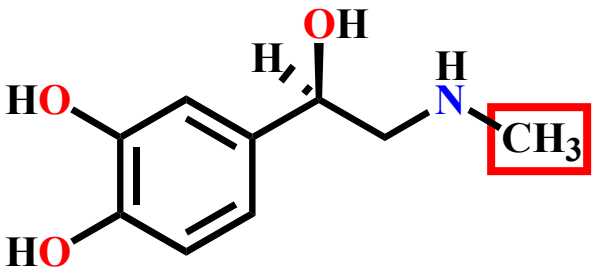


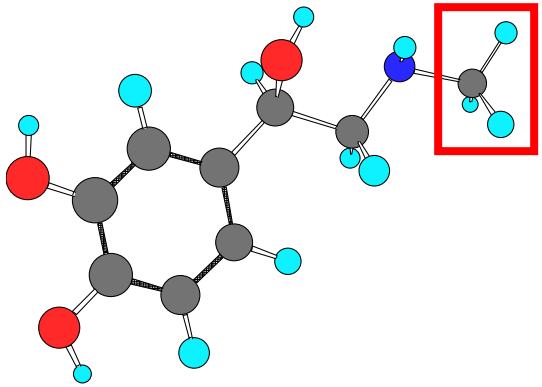
2-To introduce selectivity

Selectivity of adrenergic agonists and antagonists for b-adrenoceptors over α-adrenoceptors

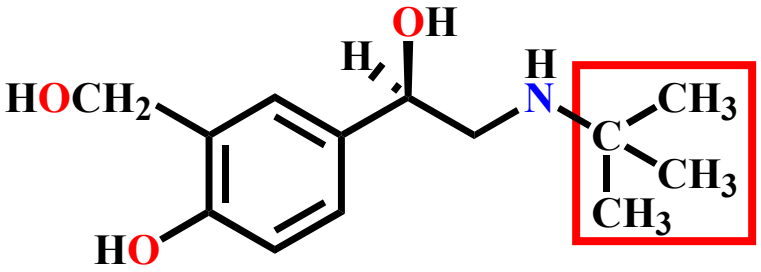


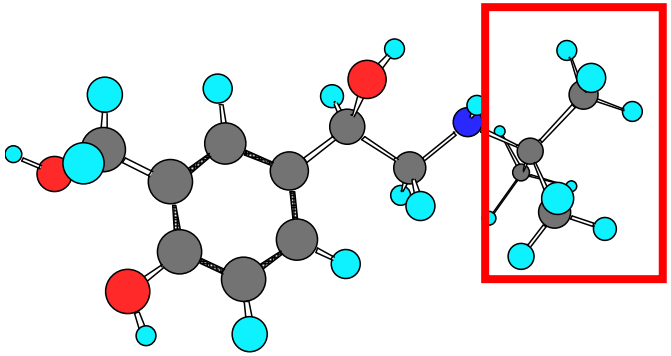
Adrenaline



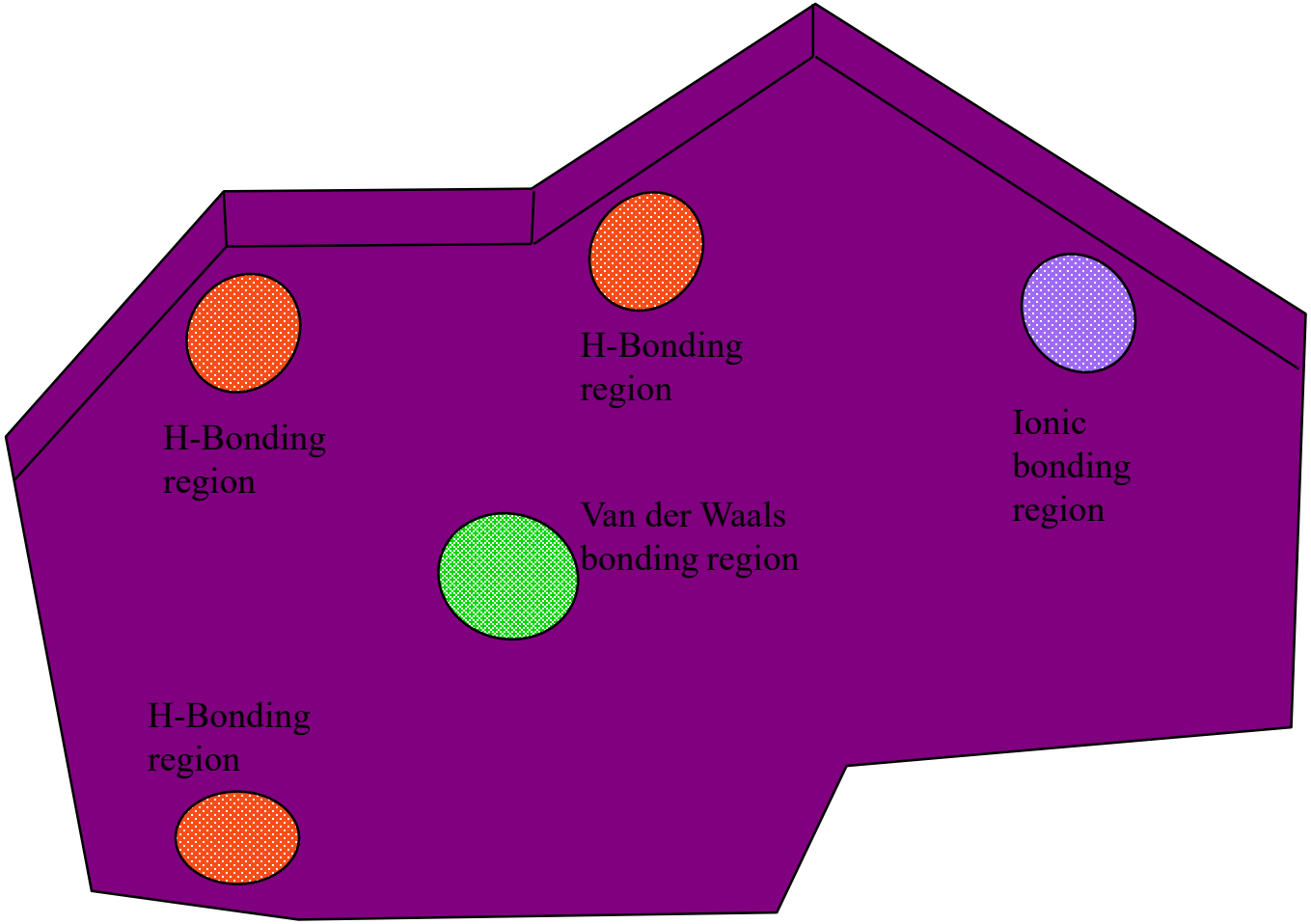


Salbutamol (Ventolin)
(Anti-asthmatic)

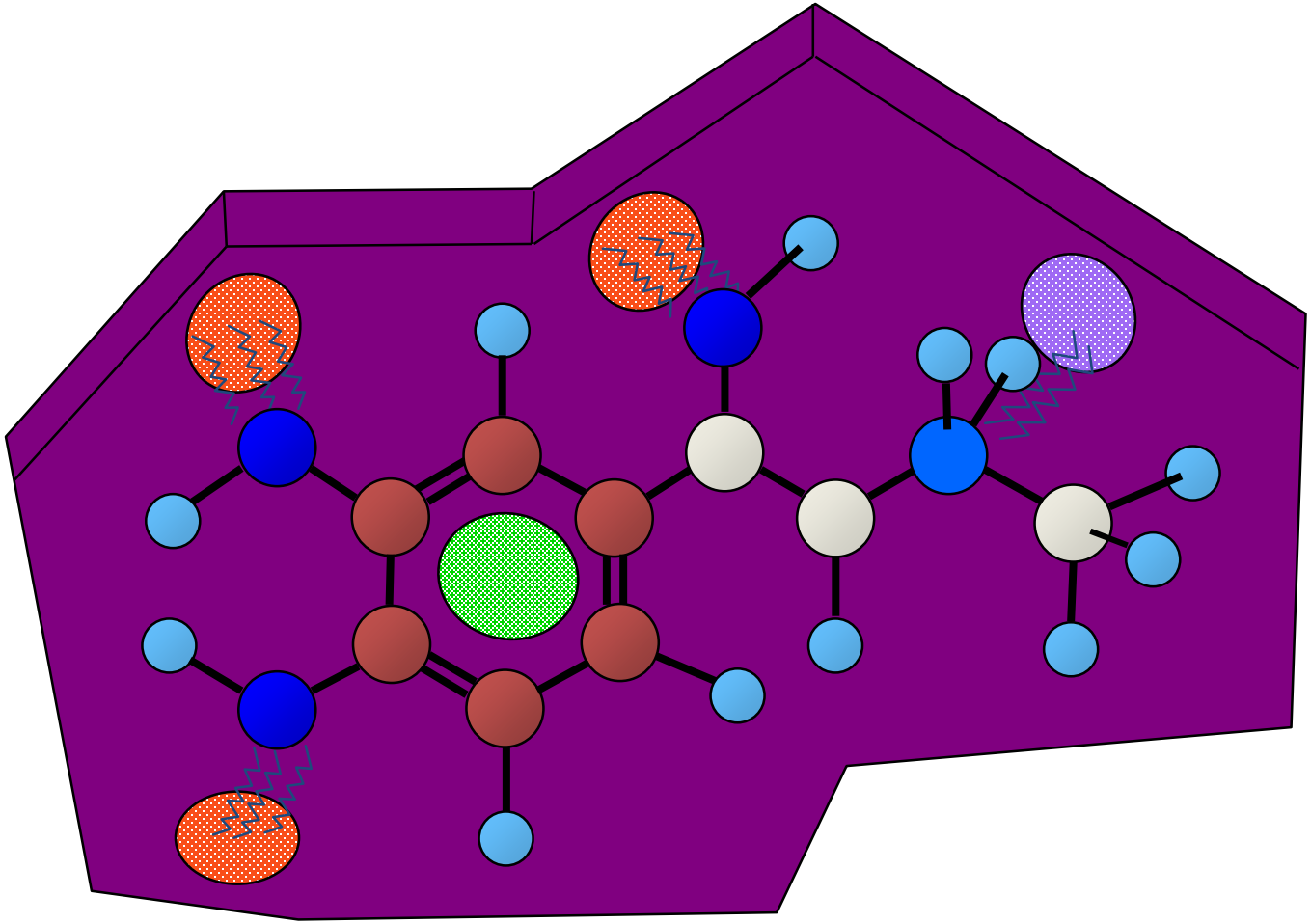




α-Adrenoceptor

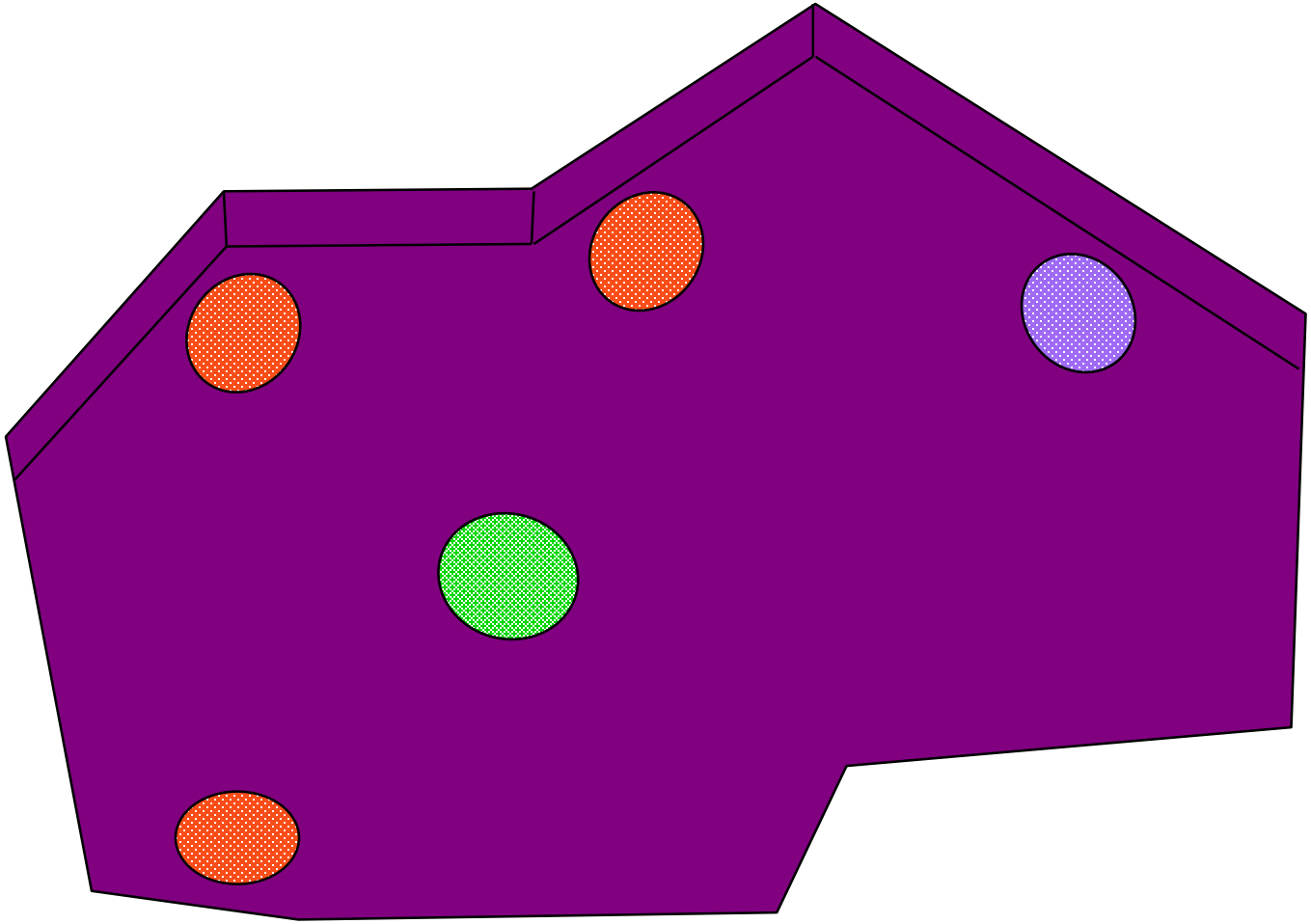


α-Adrenoceptor

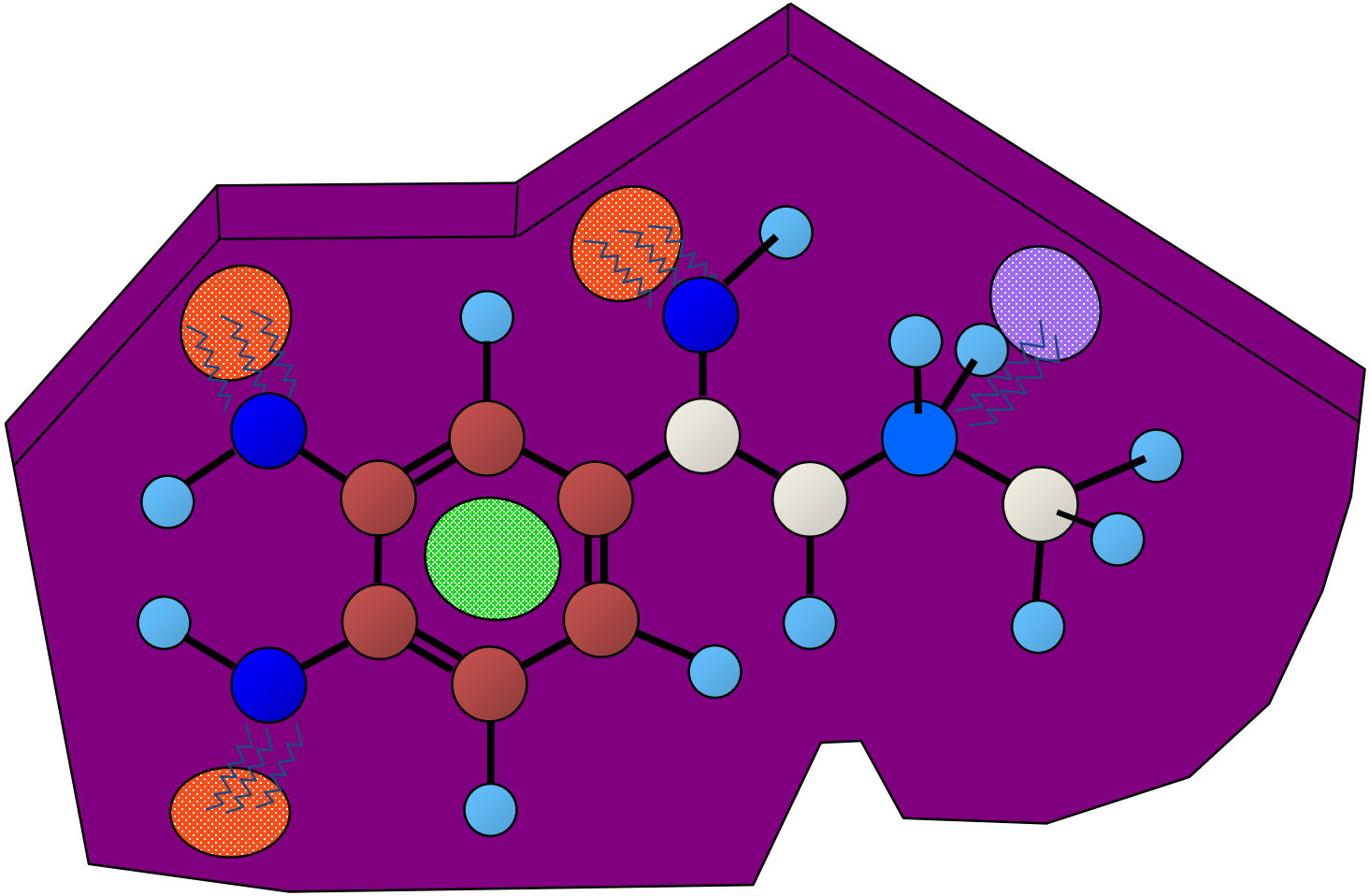


ADRENALINE

α-Adrenoceptor

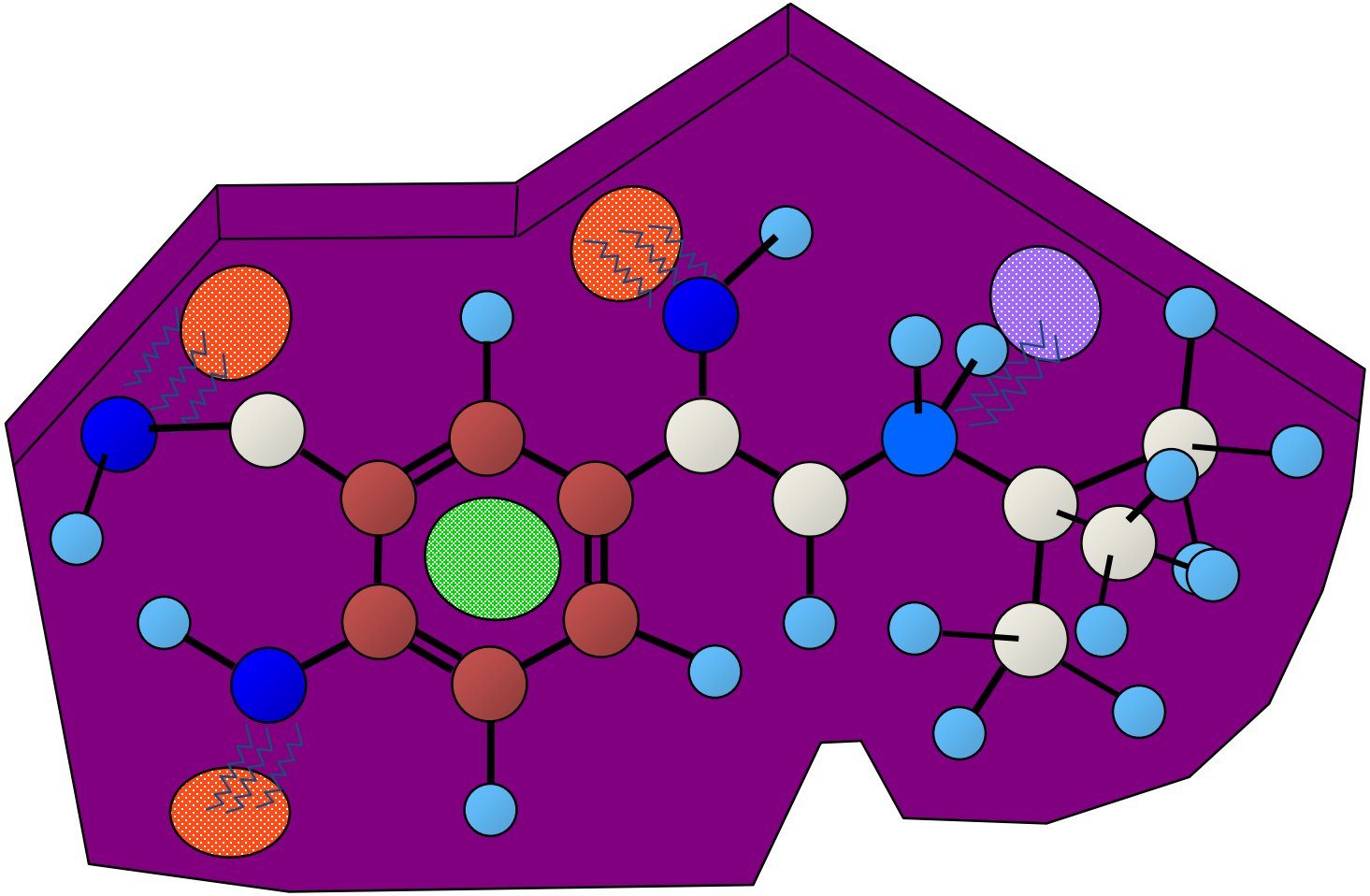


β-Adrenoceptor



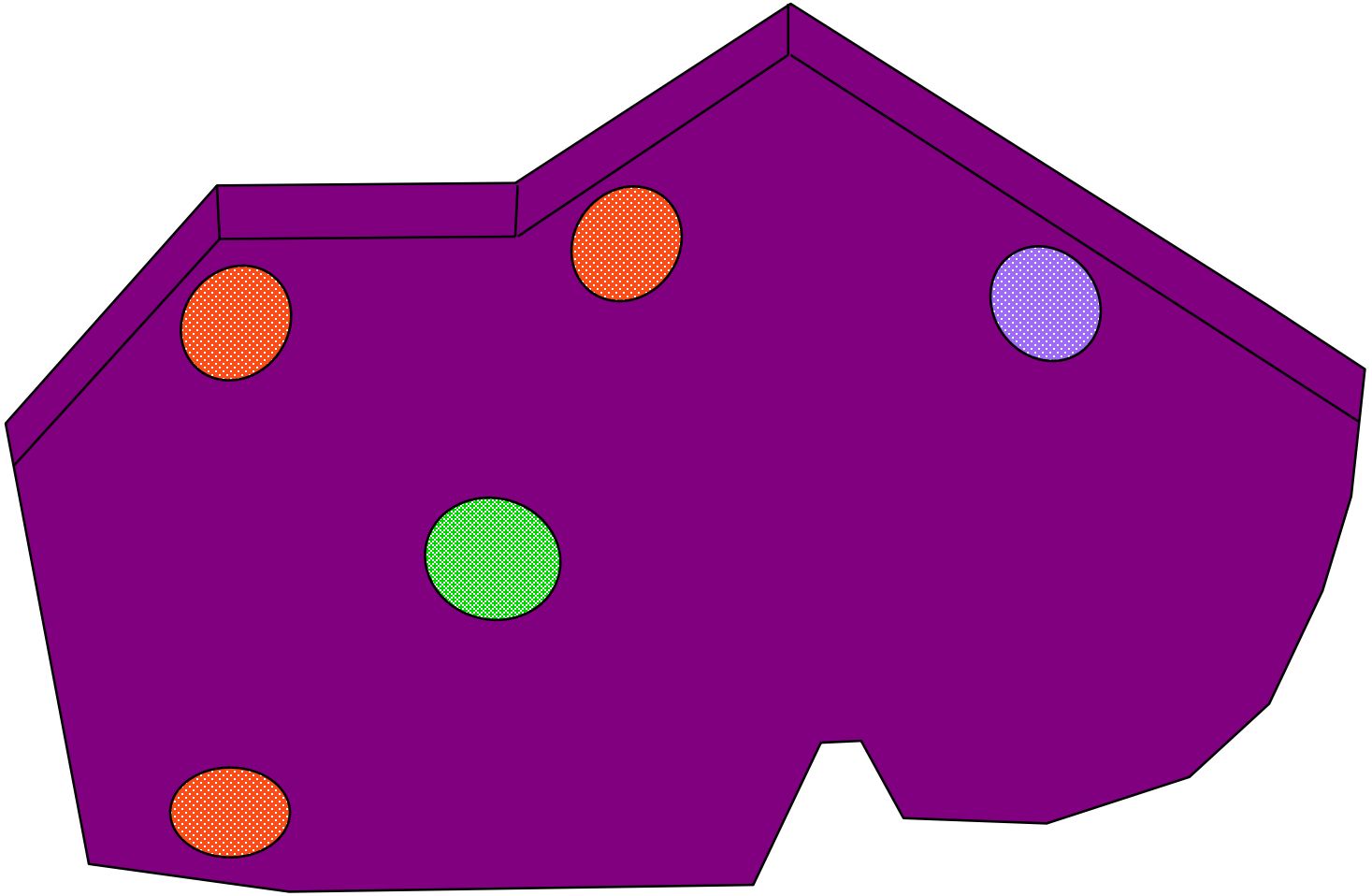
ADRENALINE

β-Adrenoceptor

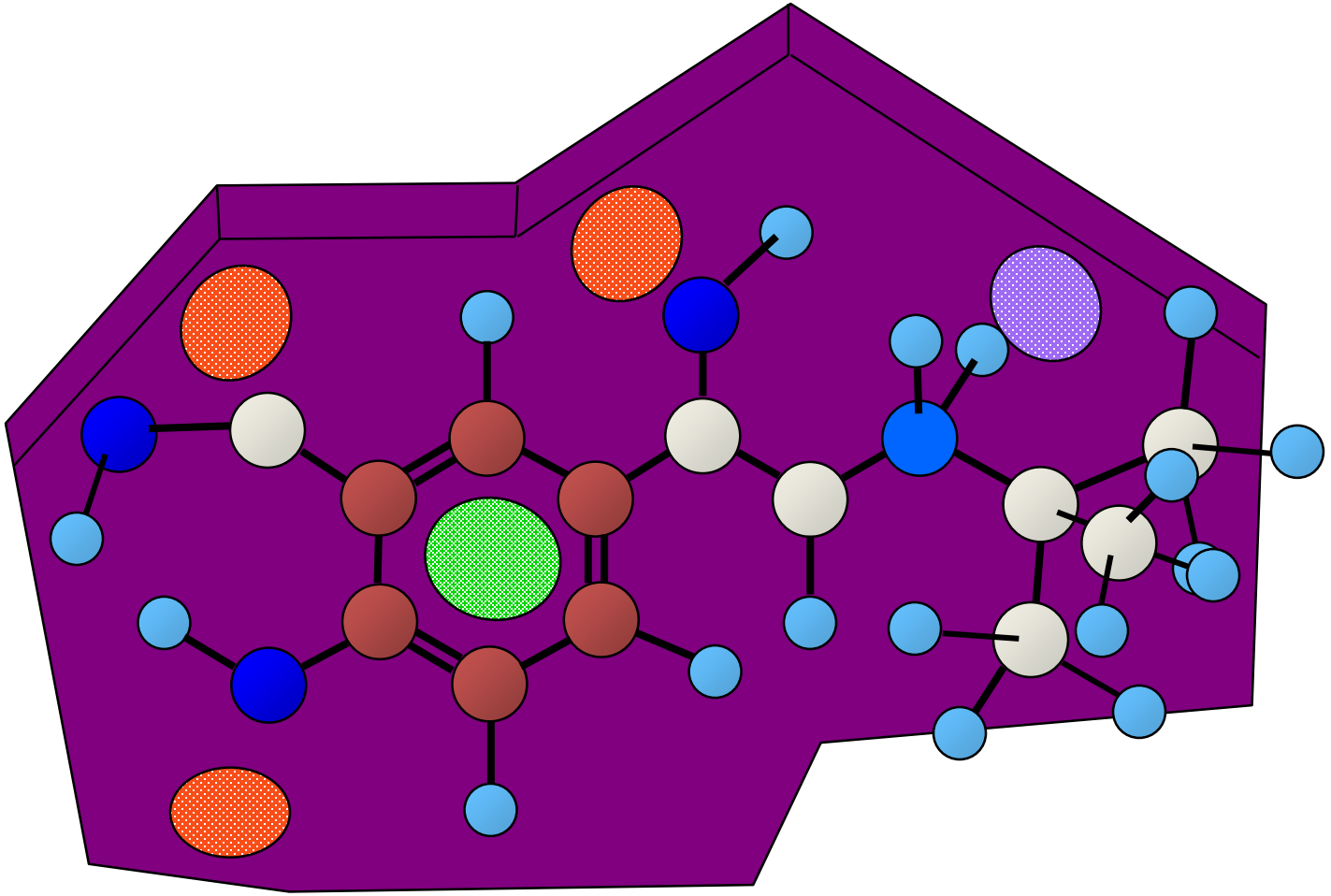


SALBUTAMOL

β-Adrenoceptor

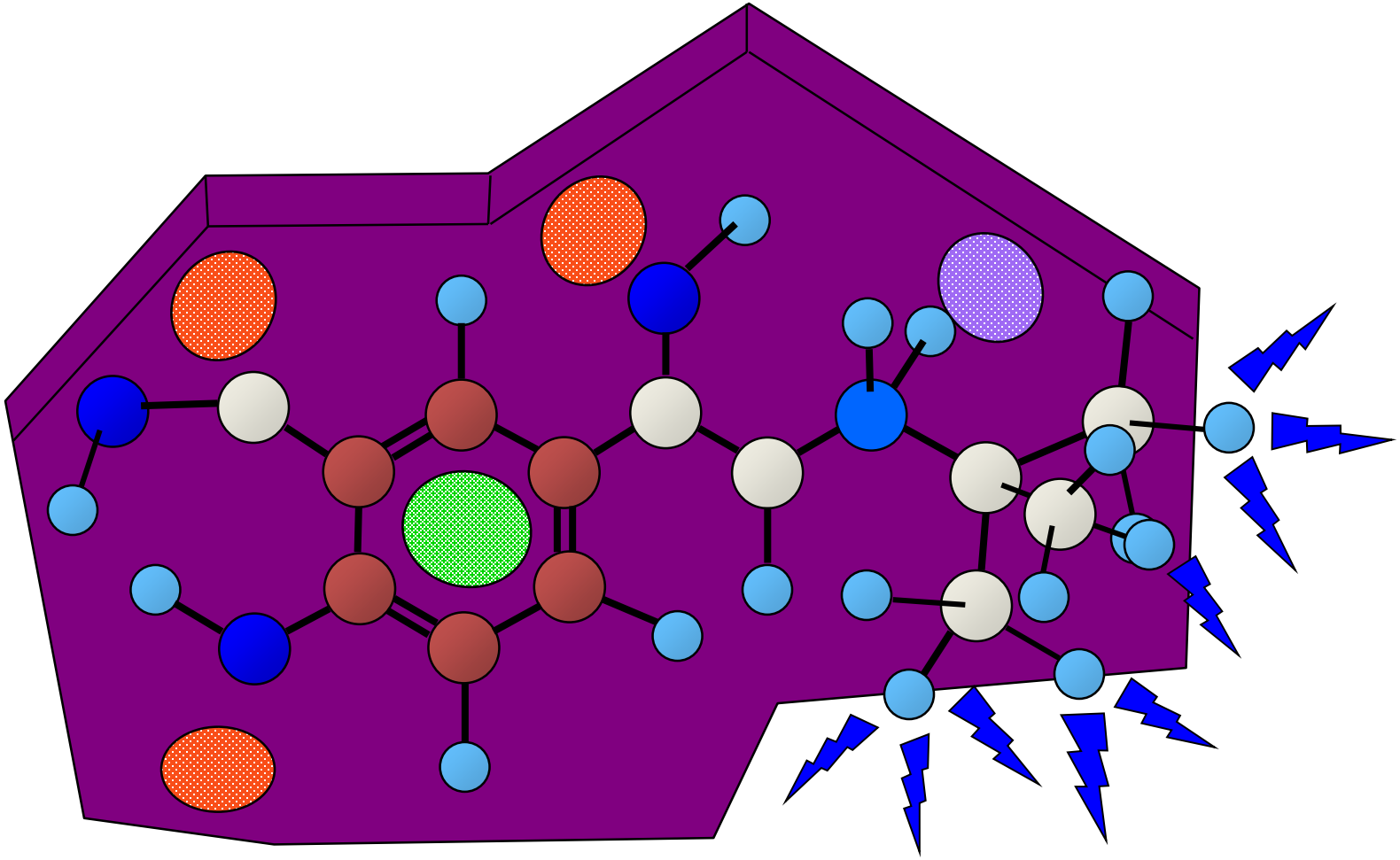


α-Adrenoceptor



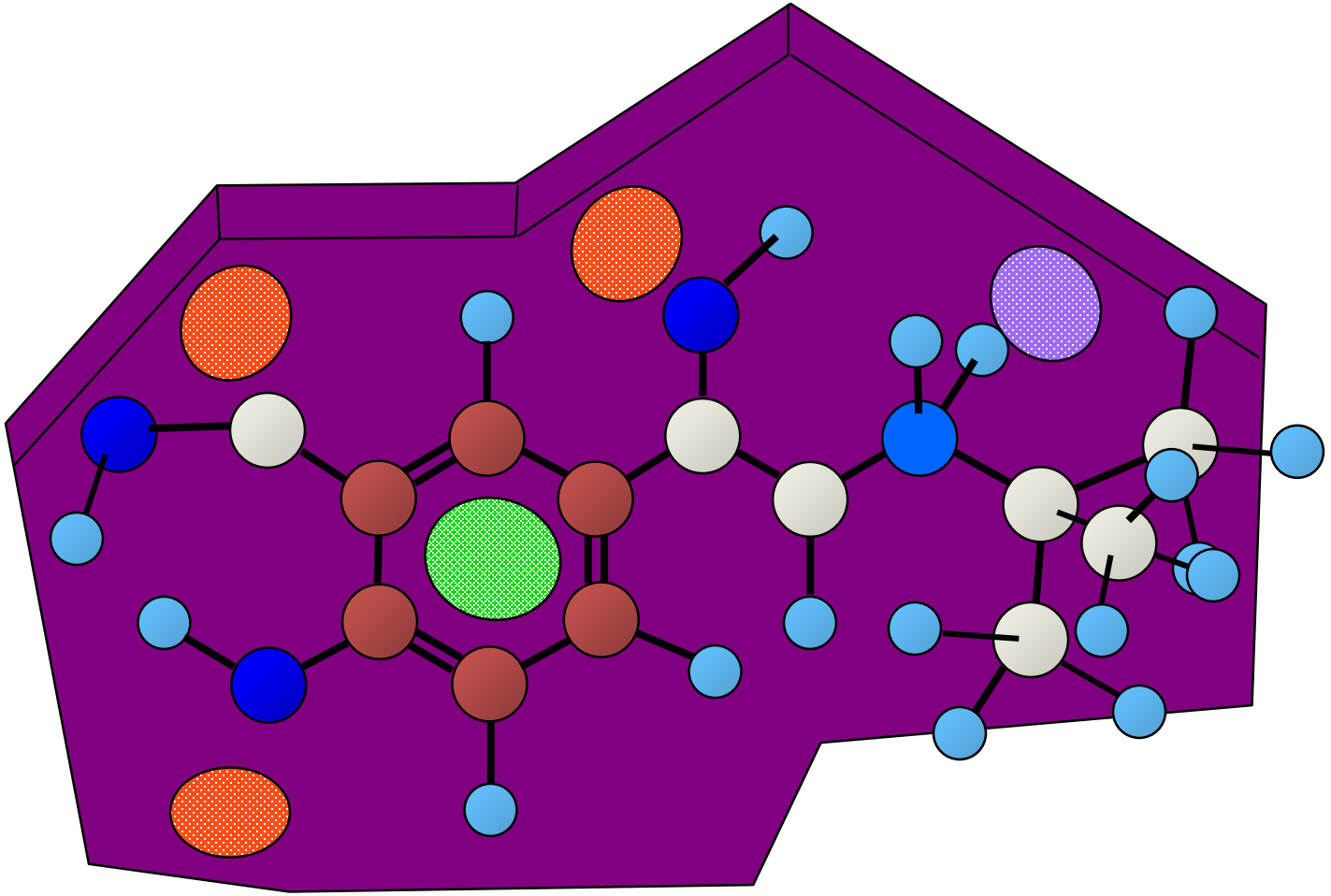
SALBUTAMOL

α-Adrenoceptor



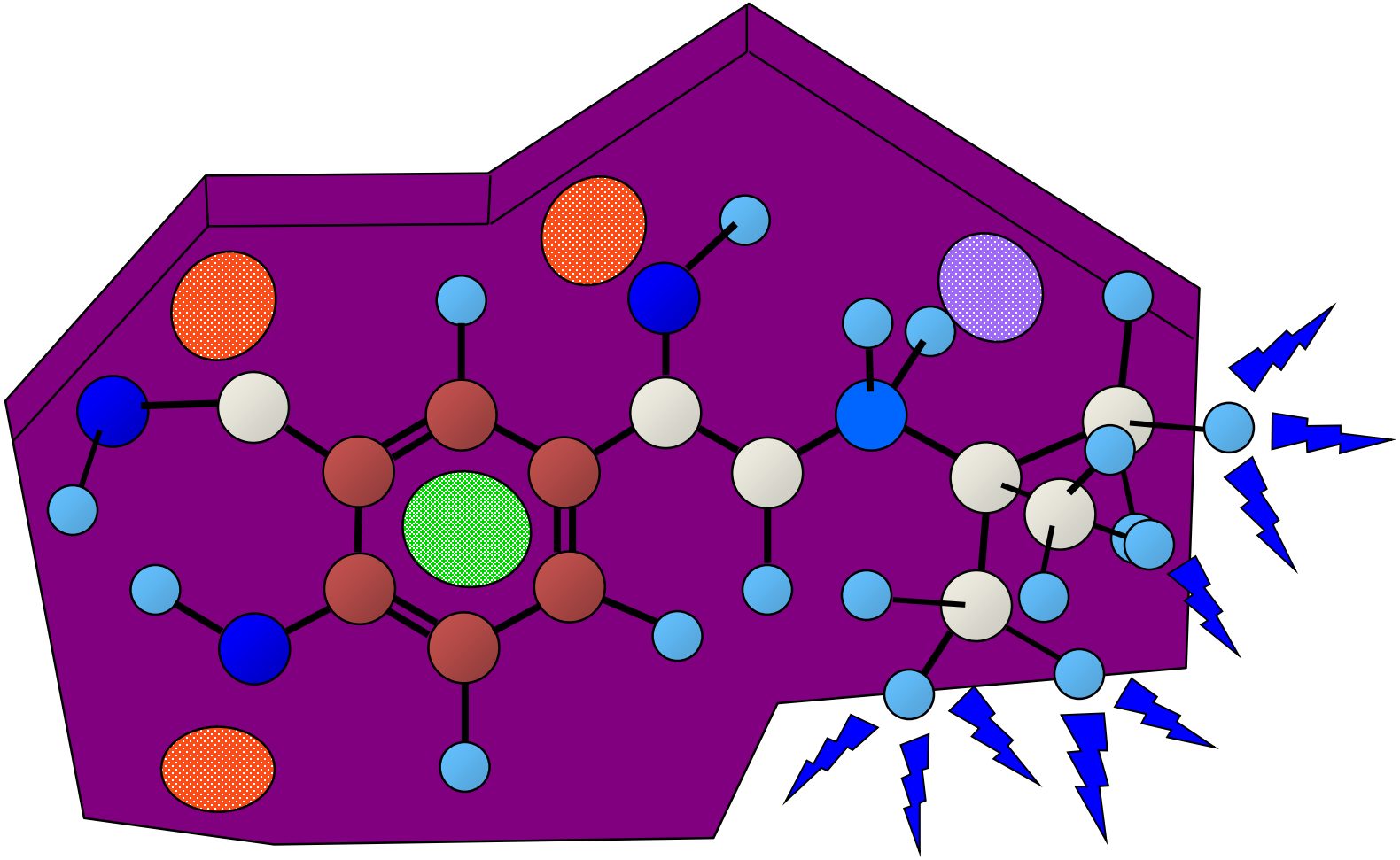
SALBUTAMOL

α-Adrenoceptor



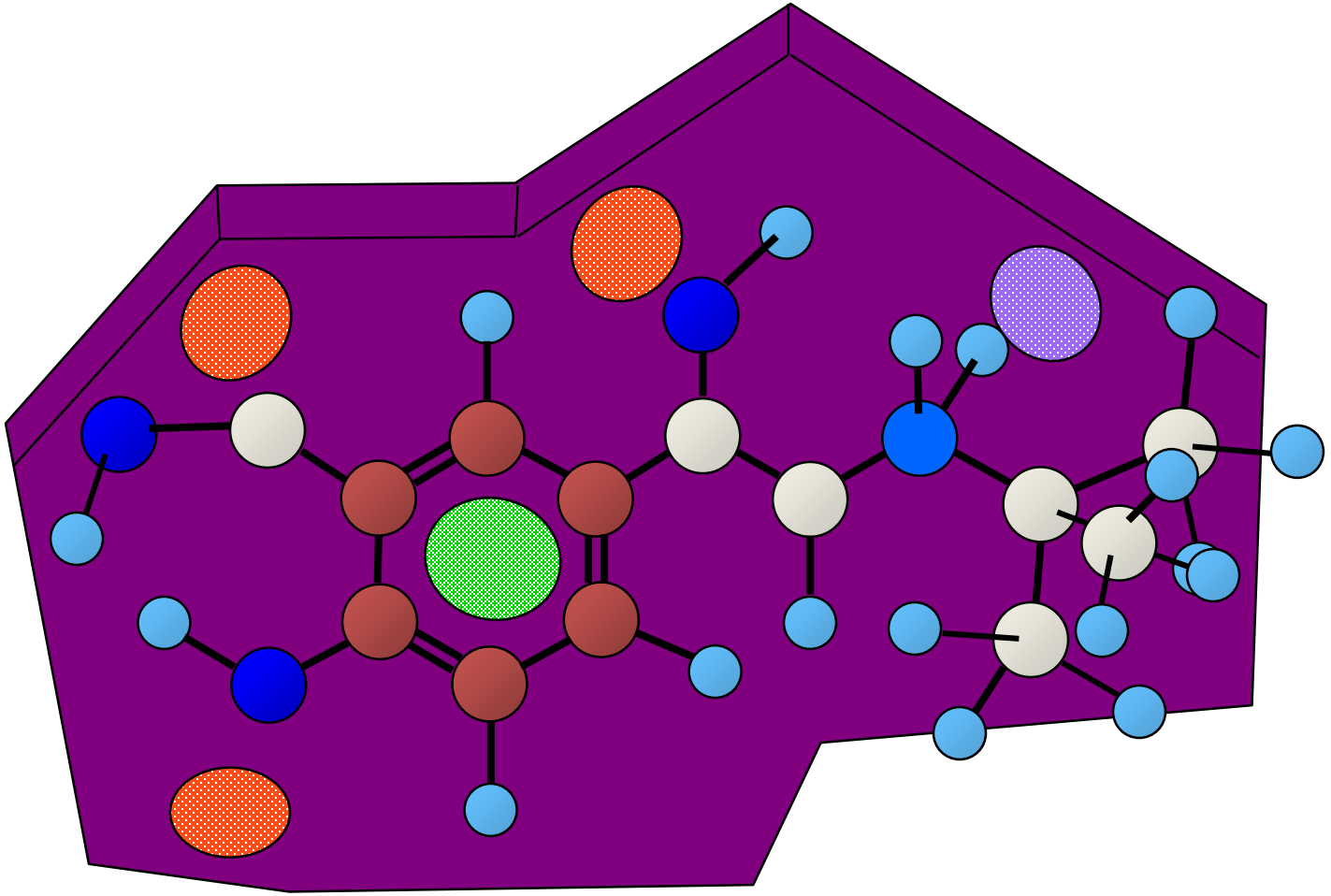
SALBUTAMOL

α-Adrenoceptor



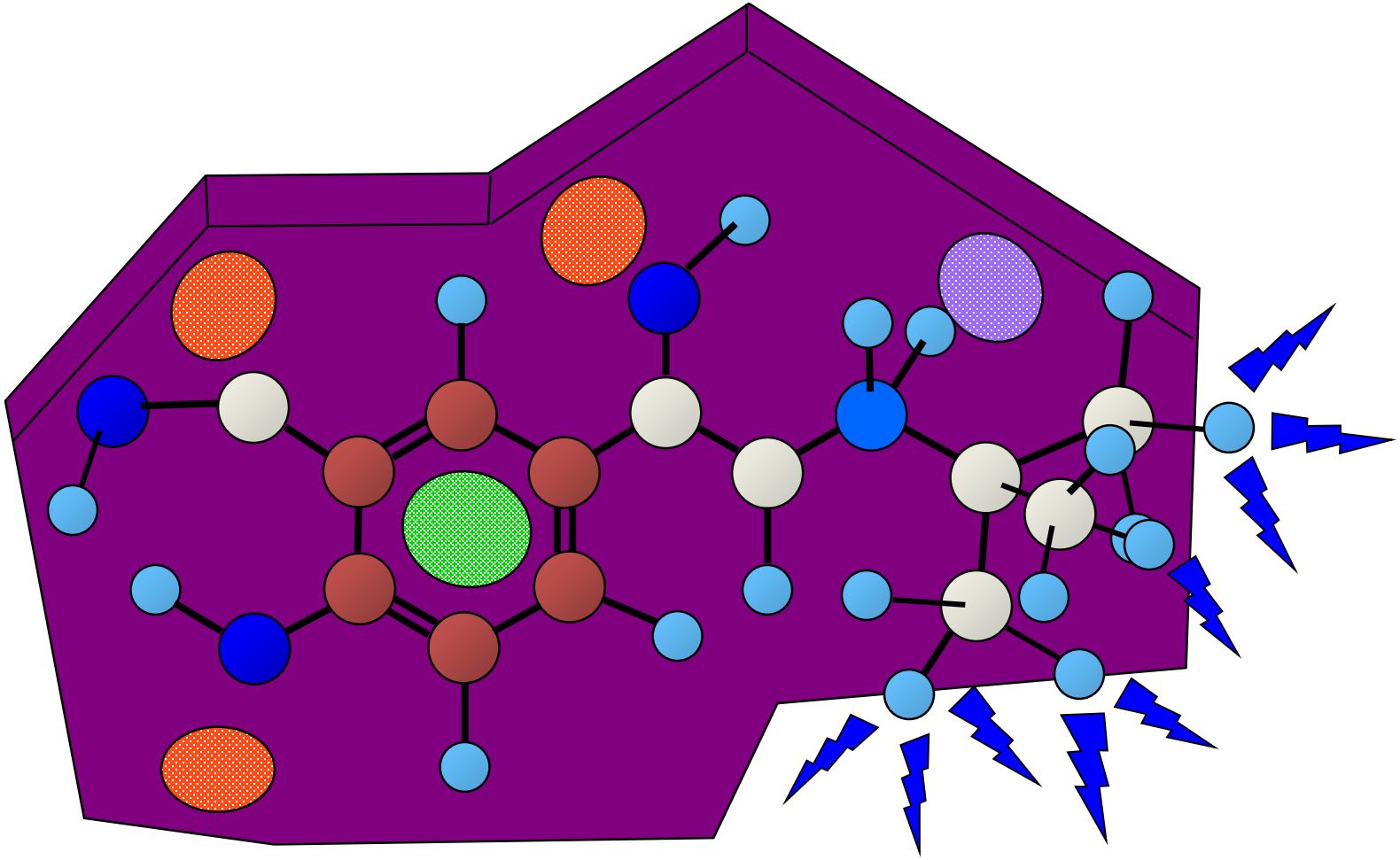
SALBUTAMOL

α-Adrenoceptor



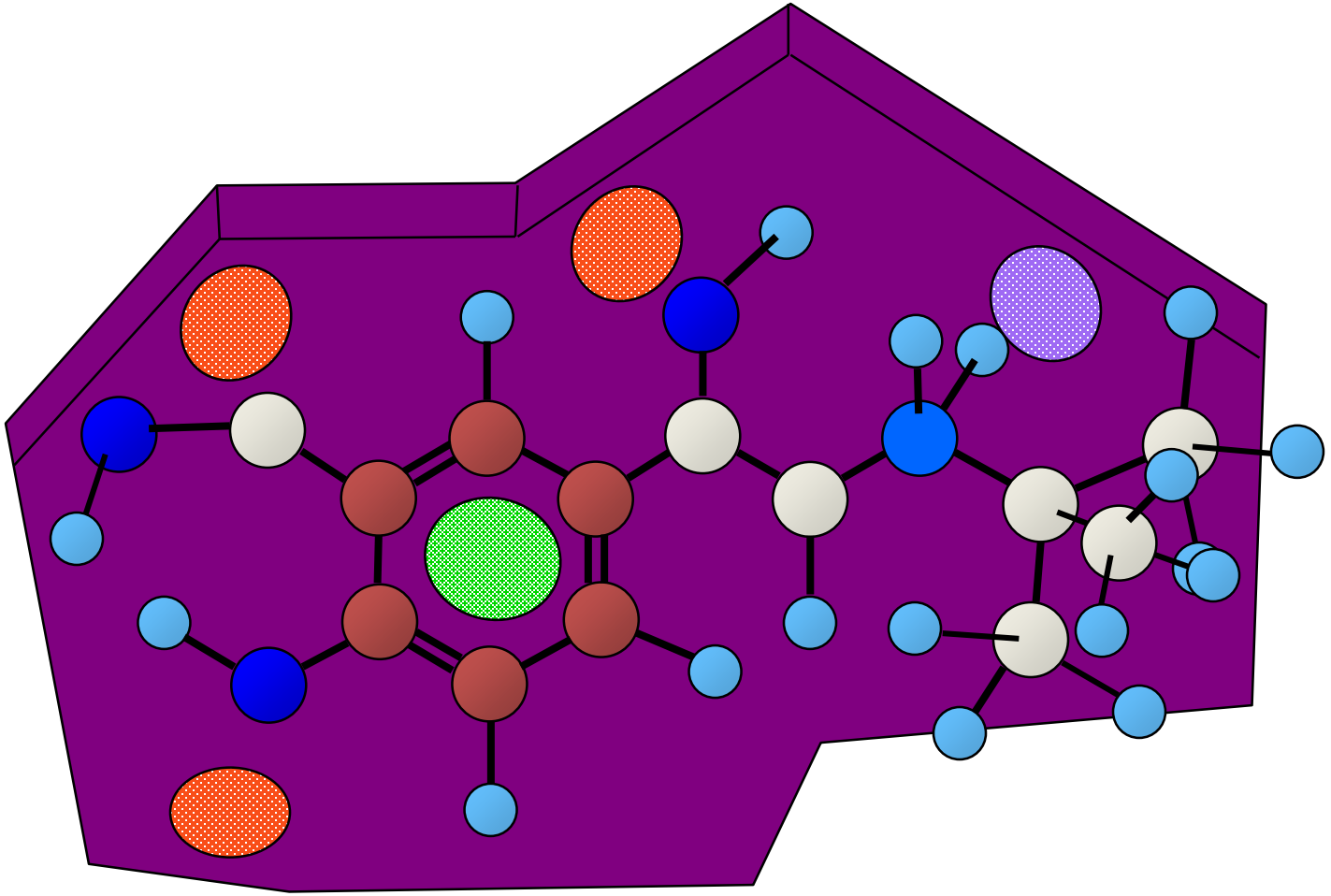
SALBUTAMOL

α-Adrenoceptor



SALBUTAMOL

α-Adrenoceptor



SALBUTAMOL

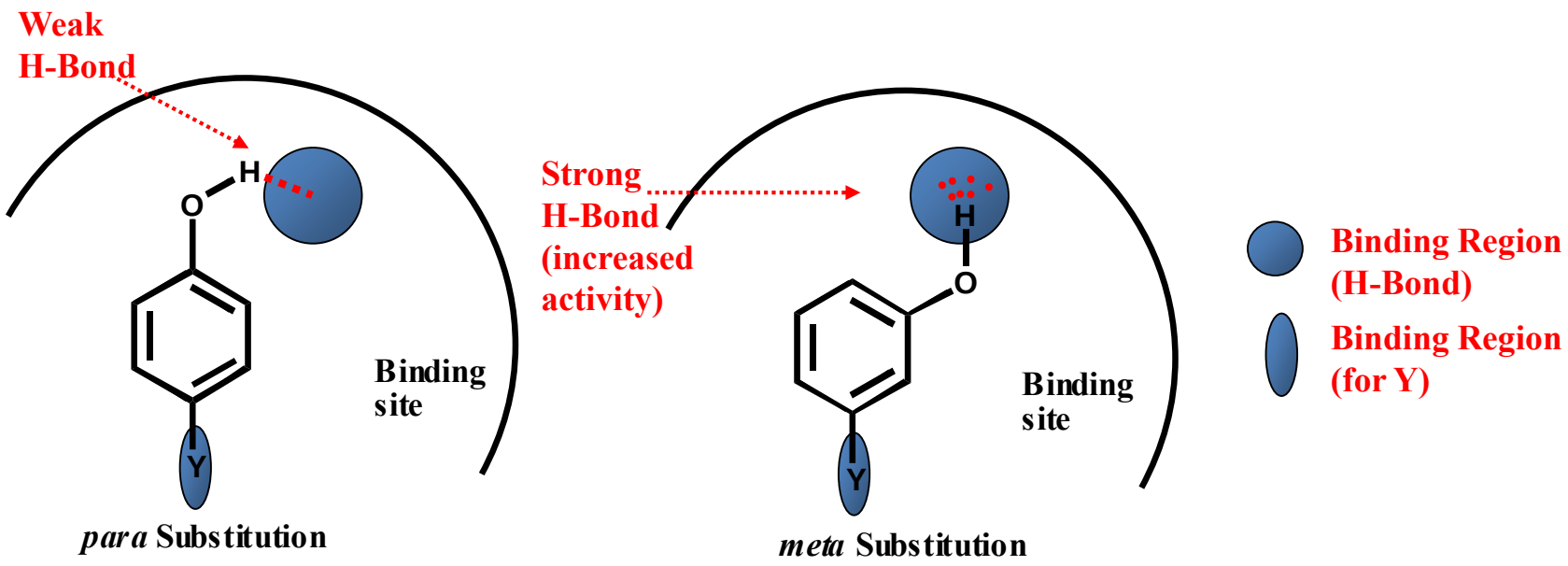
Notes on synthetic feasibility of analogues

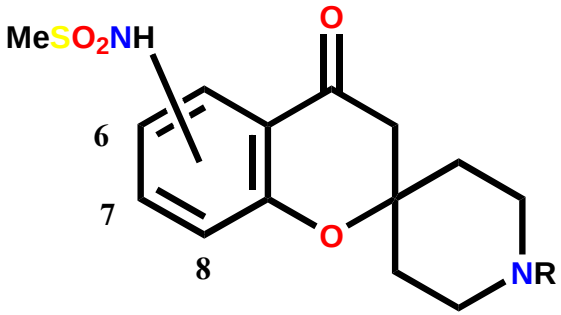
- Feasible to remove alkyl substituents on heteroatoms and replace with other alkyl substituents

- Difficult to modify alkyl substituents on the carbon skeleton of a lead compound. Full synthesis is usually required

2-Vary aryl substituents

1-Vary substitution pattern to enhance binding interactions

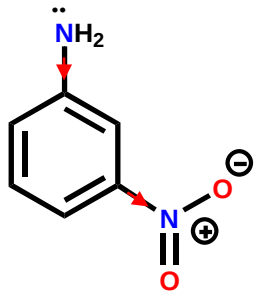




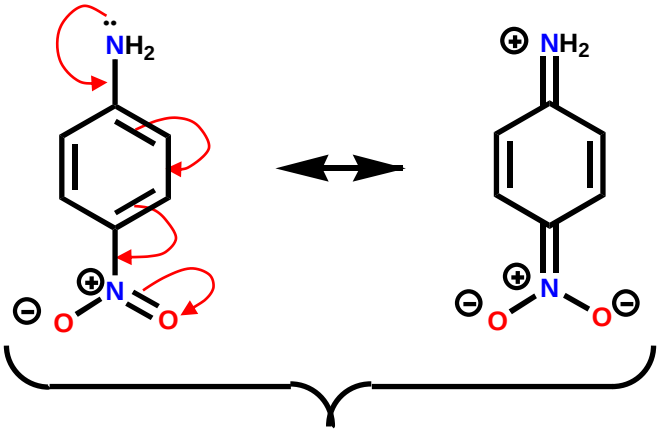
Benzopyrans

Anti-arrhythmic activity best when substituent is at 7-position

2-To enhance binding strength indirectly(electronic effects)



Meta (Inductive electron withdrawing effect)



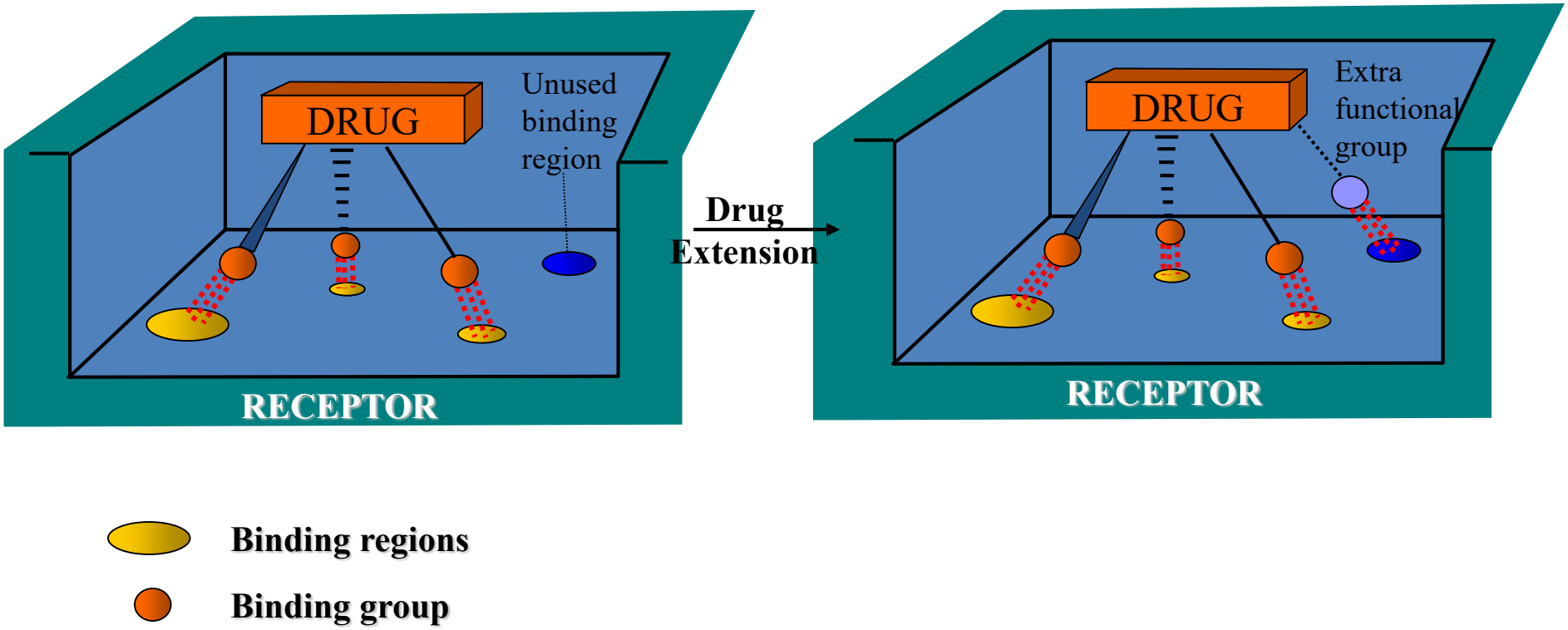
Para (Electron withdrawing effect due to resonance and inductive effects leading to a weaker base)

Vary substituents

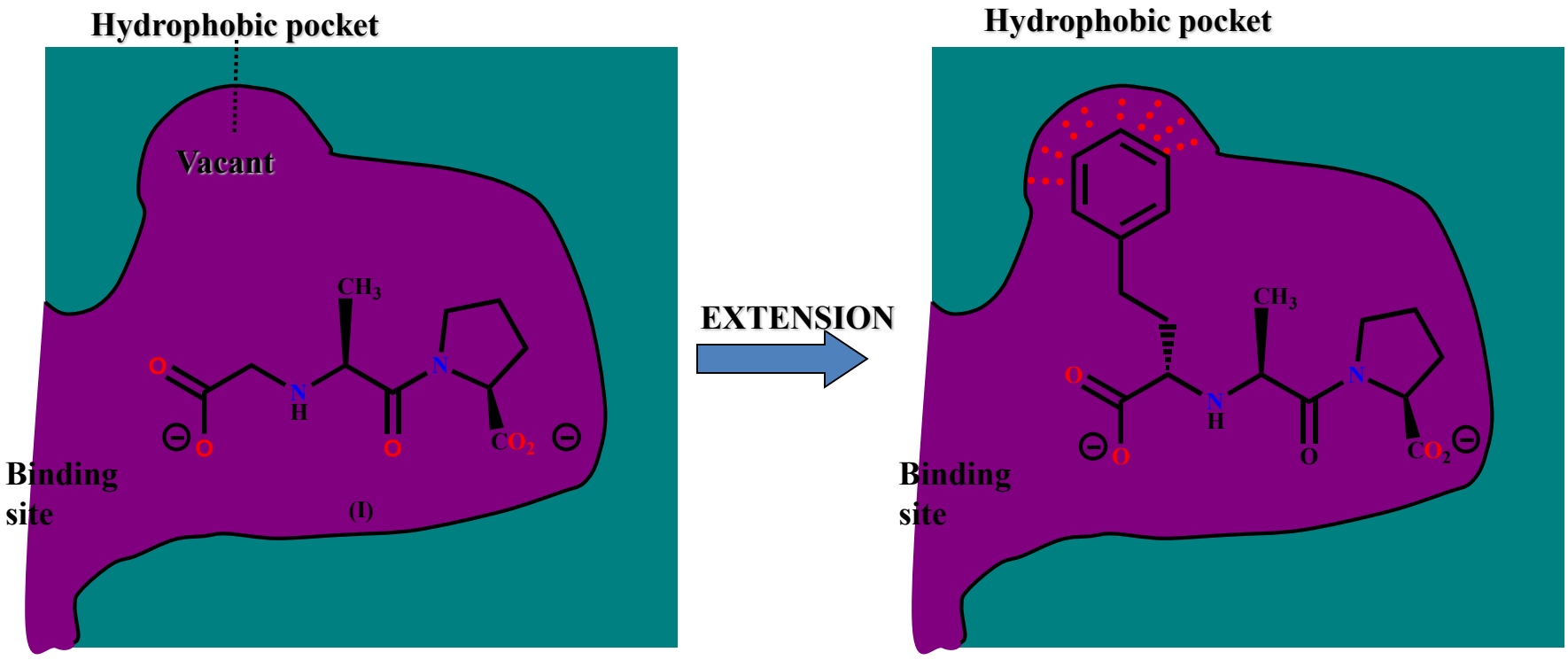
If the substitution pattern is ideal, then we can try varying the substituents themselves.

3-Chain Extension / Contraction

1-To explore target binding site for further binding regions to achieve additional binding interactions



Example : ACE Inhibitors



Lead compound

enaliprilate
1000 fold increase in activity

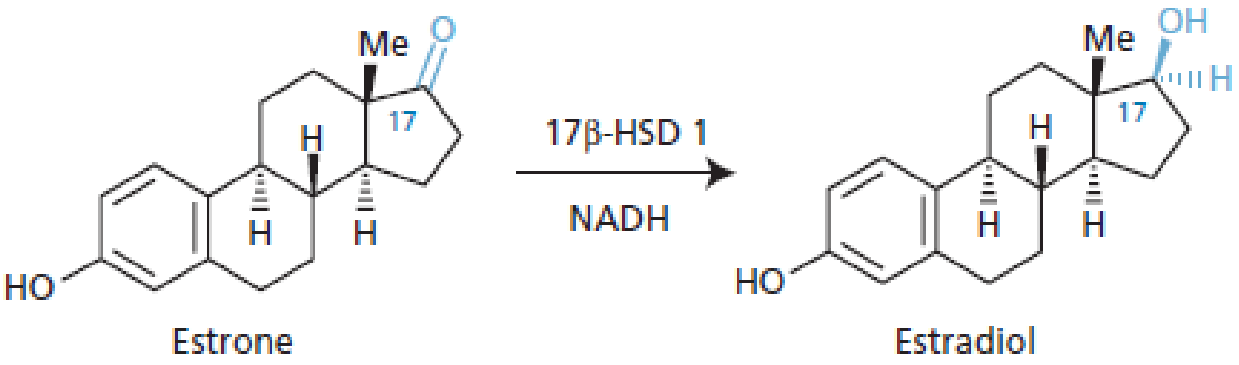


FIGURE 1 The enzyme-catalysed conversion of estrone to estradiol.

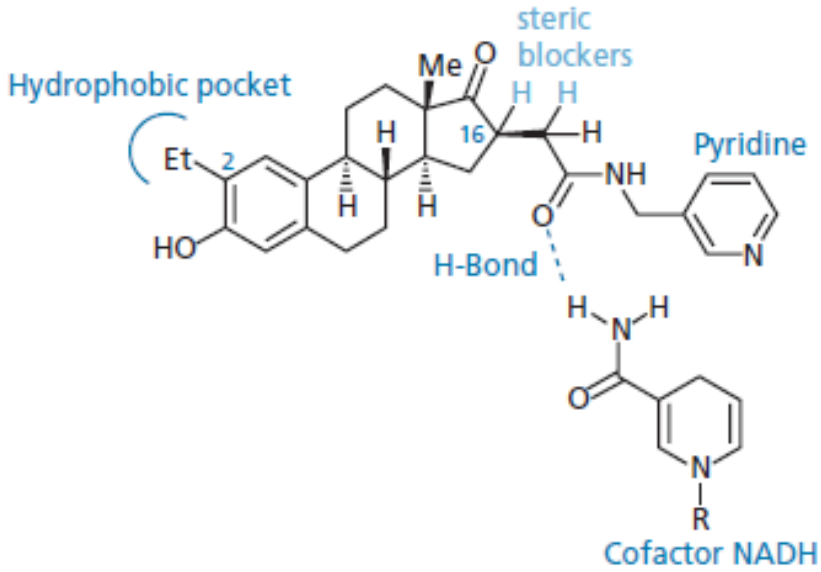
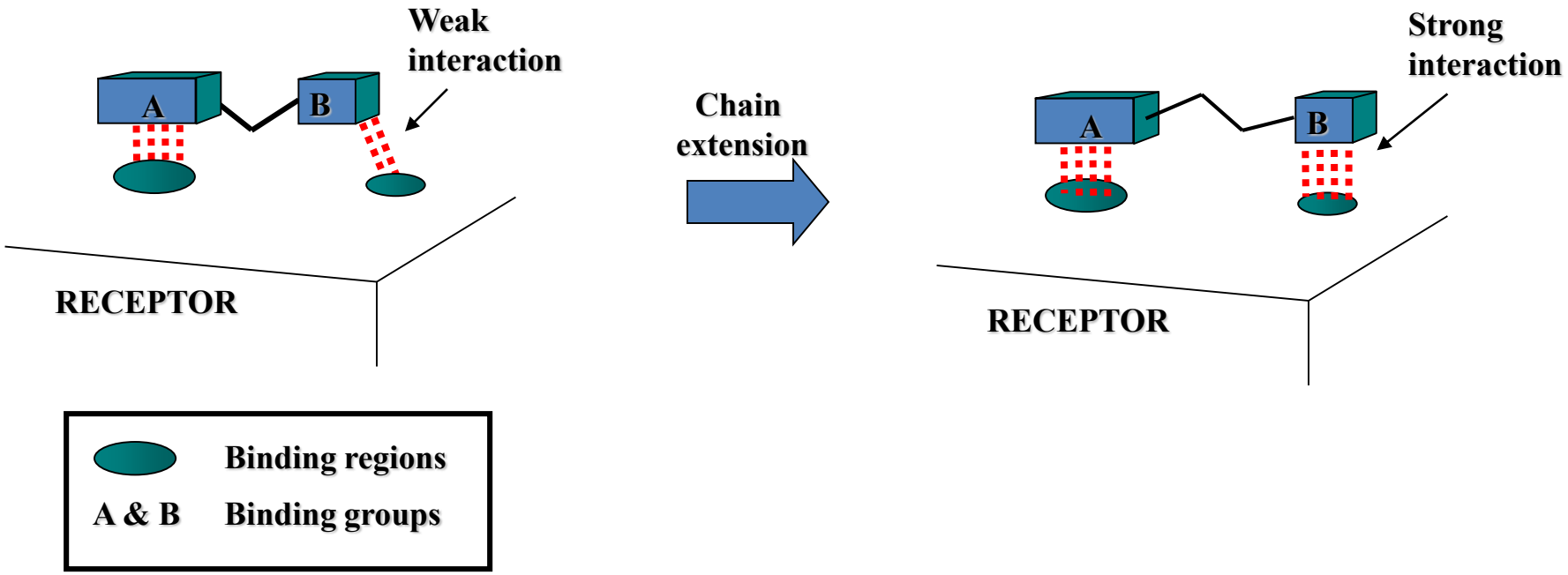


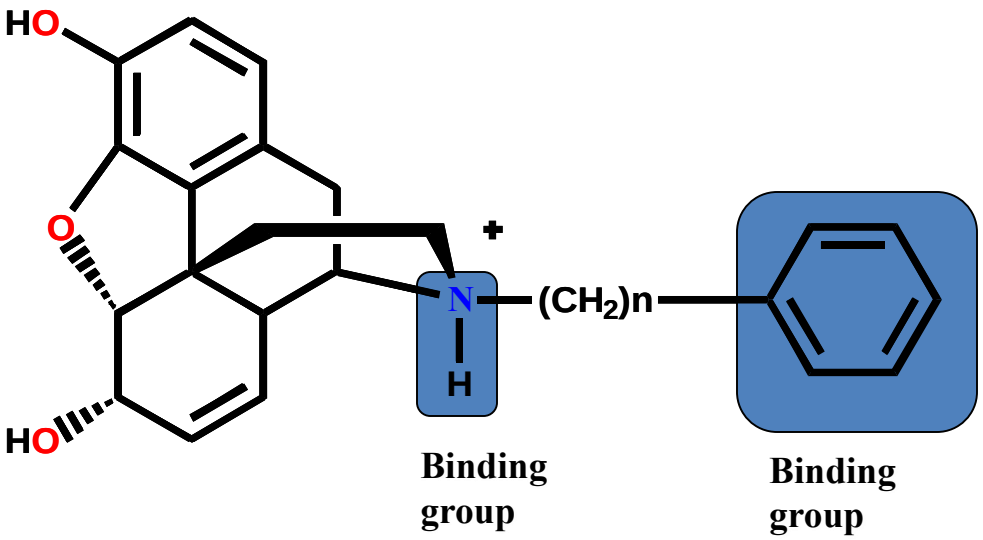
FIGURE 2 Extra binding interactions resulting from the extension strategy.

2- Vary length of chain to optimise interactions

Useful if a chain is present connecting two binding groups

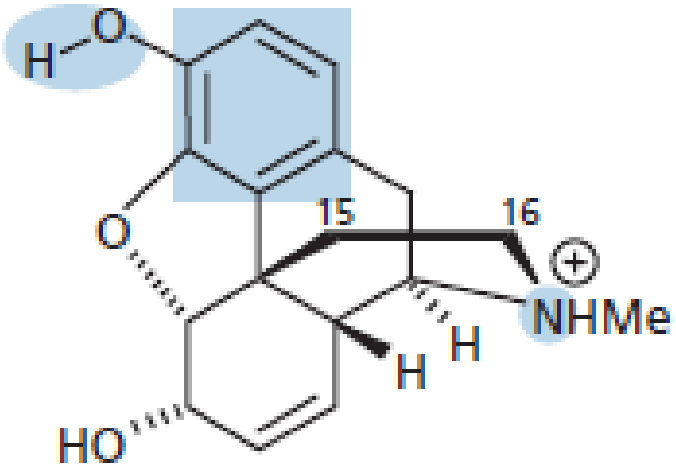


Example : *N*-Phenethylmorphine



Optimum chain length = 2

14 fold increase in activity

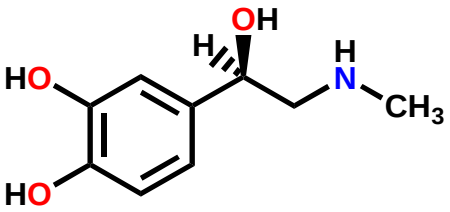


morphine

R^2 =	Me Et Pr	Bu	Pentyl, Hexyl	CH_2CH_2Ph
	Agonism decreases Antagonism increases	Zero activity	Agonists	14 × activity of morphine

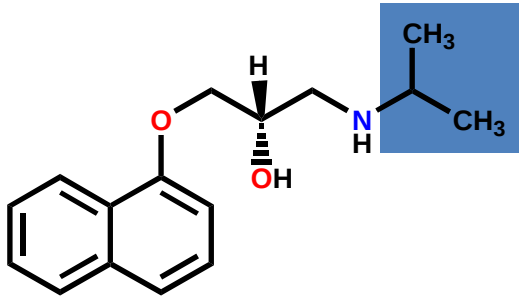
FIGURE 24.10 Change in activity with respect to alkyl group size.

3- Antagonists from agonists

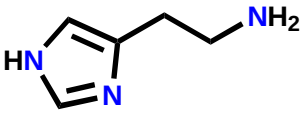


Adrenaline



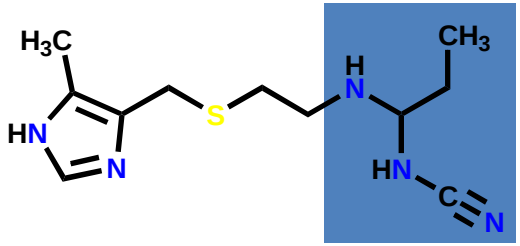


Propranolol
(β-Blocker)



Histamine

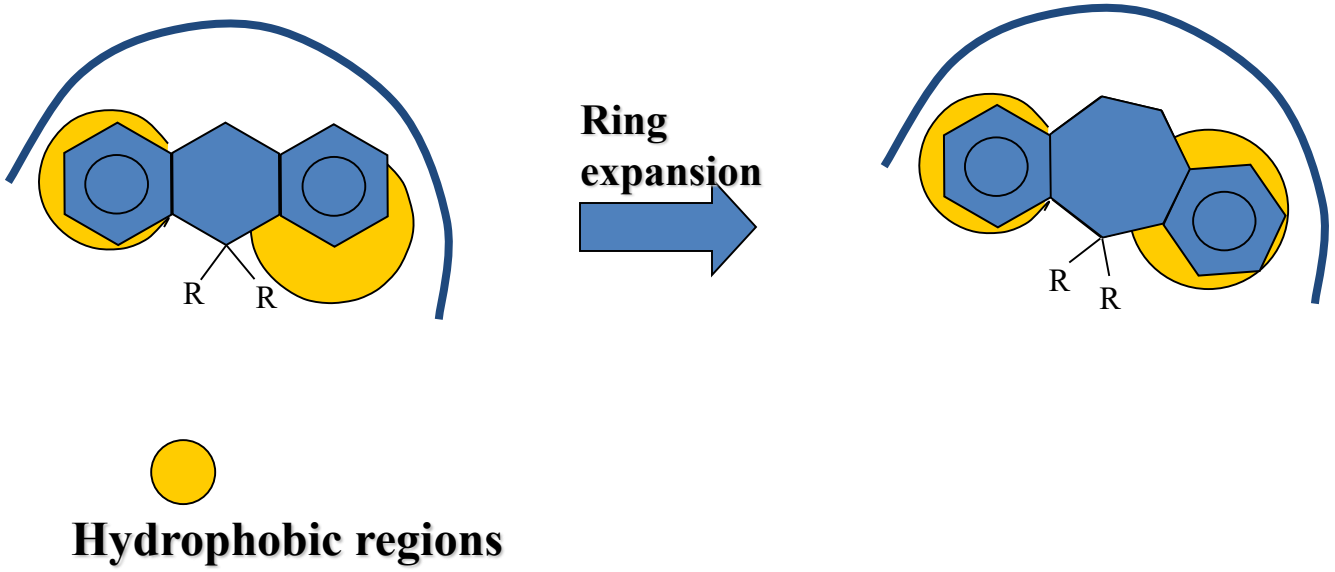


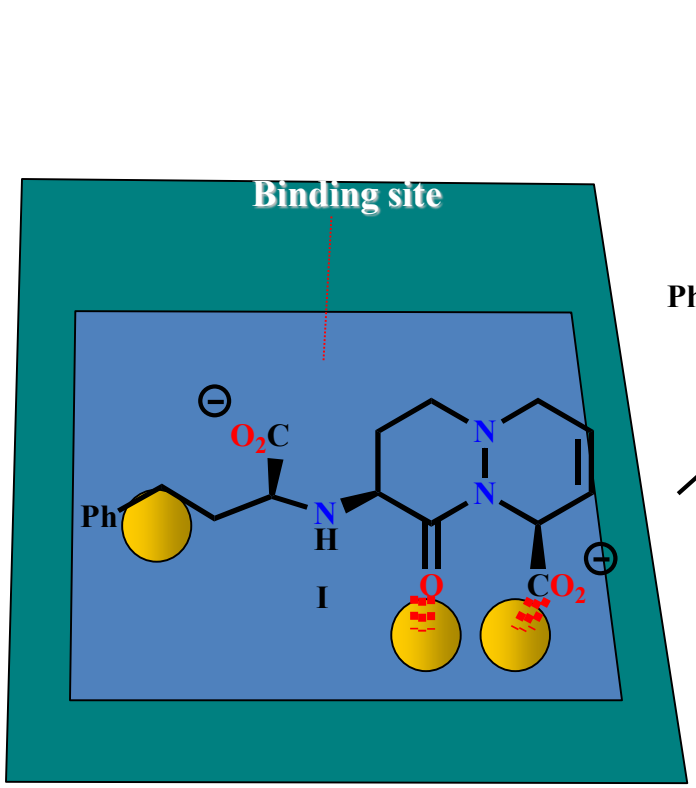


Cimetidine (Tagamet)
(Anti-ulcer)

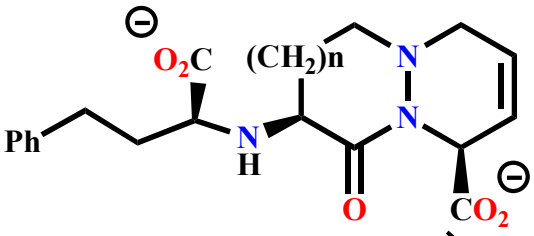
4-Ring Expansion / Contraction

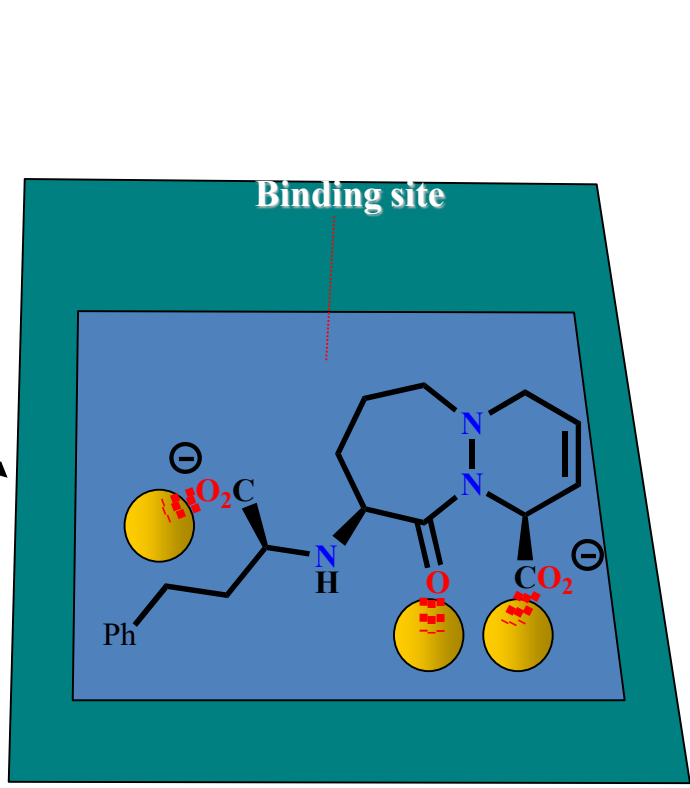
1-To improve overlap of binding groups with their binding regions





Vary n to vary ring size





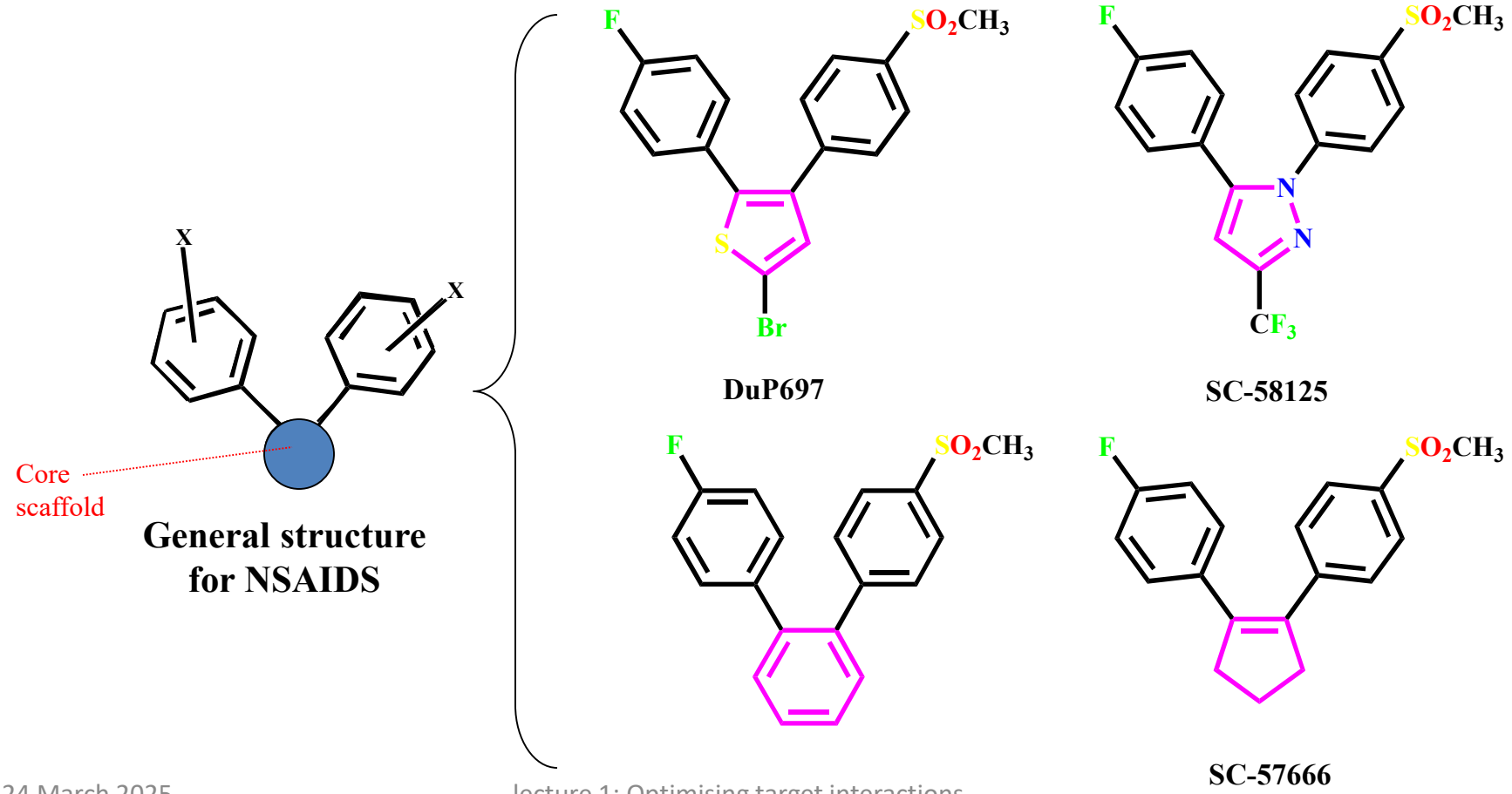
Binding regions

cilazaprilat
(ACE inhibitor)

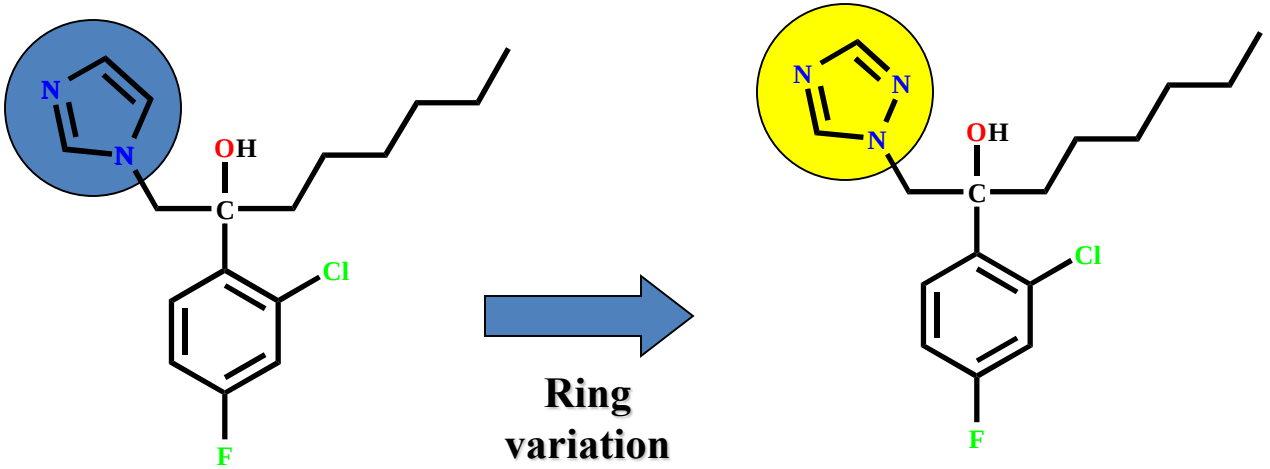
5-Ring Variation

1-Often done for patent reasons

Replace aromatic/heterocyclic rings with other ring systems



2-Sometimes results in improved selectivity



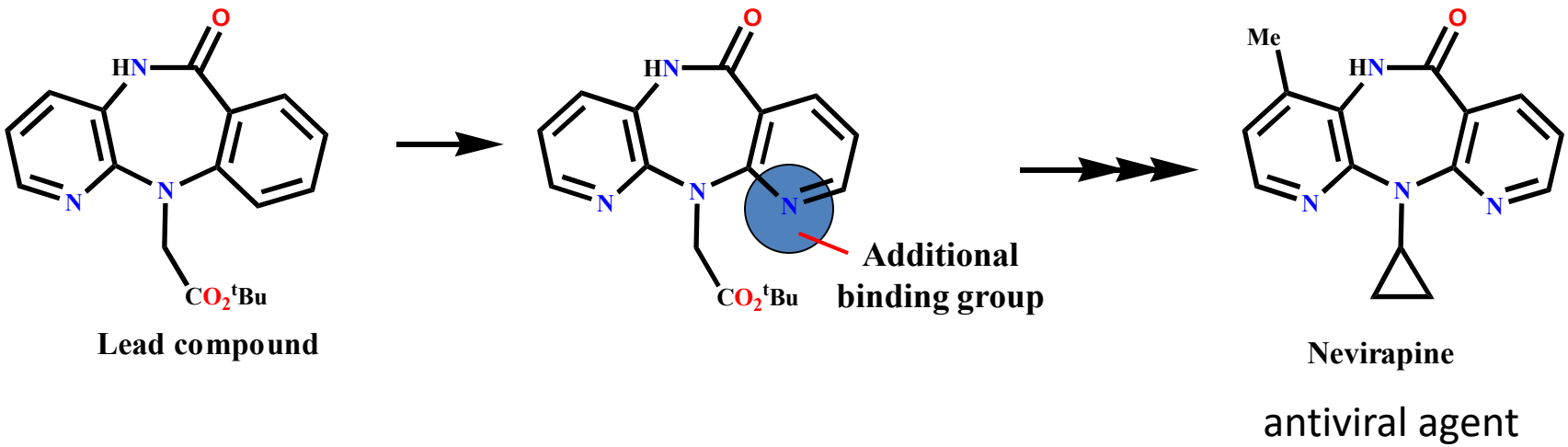
Structure I

Antifungal agent

UK-46245

Improved selectivity vs. fungal enzyme

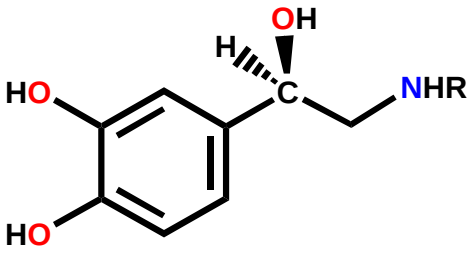
3- Introduces the possibility of an extra hydrogen bonding interaction with the binding site, should a suitable binding region be available



6-Ring fusions

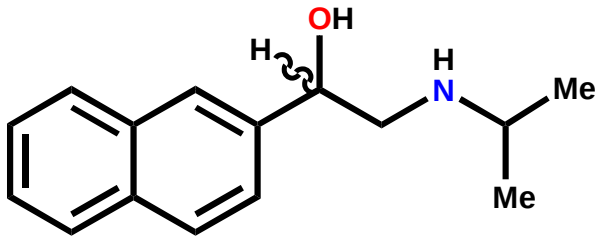
1- increased interactions or increased selectivity

Pronethalol (β-blocker)



R = Me Adrenaline
R = H Noradrenaline





Pronethalol

Selective for b-receptorsover α-receptors

- larger van der Waals binding area
- Naphthalene ring system is sterically too big for the α-receptor

7-Isosterism

Isosteres : are atoms or groups of atoms which share the same valency and which have chemical or physical similarities

Non-classical isosteres: are groups that do not obey the steric and electronic rules used to define classical isosteres, but which have similar physical and chemical properties.

Bioisostere is used in drug design and includes both classical and non-classical isosteres. A bioisostere is a group that can be used to replace another group while retaining the desired biological activity

Bioisosterism: Classification

I. Classic isosteres:

A. Monovalent atoms or groups

B. Divalent atoms or groups

C. Trivalent atoms or groups

D. Tetravalent atoms

E. Ring equivalents

II. Non-classical isosteres:

A. Cyclic vs. Non cyclic

B. Non classic bioisosterism of functional groups

B.1. Carboxylic group bioisosteres

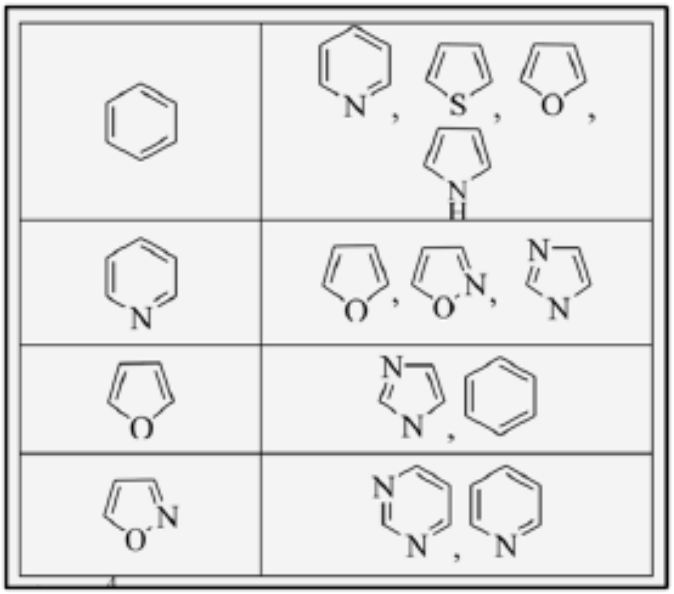
B.2 Hydroxyl group bioisosteres

B.3. Amide group bioisosteres

B.4. Halogen bioisosteres.

Classic Bioisosteres:

Monovalent	Divalent	Trivalent	Tetravalent
—OH, —NH_2, —CH_3, —OR	—CH_2—	=CH—	=C=
—F, —Cl, —Br, —I, —SH, —PH_2	—O—	=N—	=Si=
—Si_3, —SR	—S—	=P—	$=N^+=$
	—Se—	=As—	$=P^+=$
	—Te—	=Sb—	$=As^+=$
			$=Sb^+=$



Non- Classic Bioisosteres

—CO—	—COOH	$—SO_2NH_2$	—H	—CONH—	—COOR	$—CONH_2$
$—CO_2—$	$—SO_3H$	$—PO(OH)NH_2$	—F	—NHCO—	—ROCO—	$—CSNH_2$
$—SO_2—$	—tetrazole					
$—SO_2NR—$	$—SO_2NHR$ $—SO_2NH_2$		—OH $—CH_2OH$		—catechol	
—CON—	—3-hydroxyisoxazole				—benzimidazole	
—CH(CN)—	—2-hydroxychromones		$—NHCONH_2$			C_4H_4S
R—S—R			$—NH—CS—NH_2$			$—C_5H_4N$
(R—O—R)	=N—					$—C_6H_5$
R—N(CN)—	C(CN)=R′		$—NH—C(=CHNO_2)—NH_2$ $—NH—C(=CHCN)—NH_2$			
—halide						$—C_4H_4NH$
	$—CF_3$					
	—CN					
	$—N(CN)_2$					
	$—C(CN)_3$					

Why isosterism

1- To determine whether a particular group is an important binding group

2- Isosteric groups could be used to determine whether a particular group is involved in hydrogen bonding.

Replacing O with CH_2…… little difference to the size

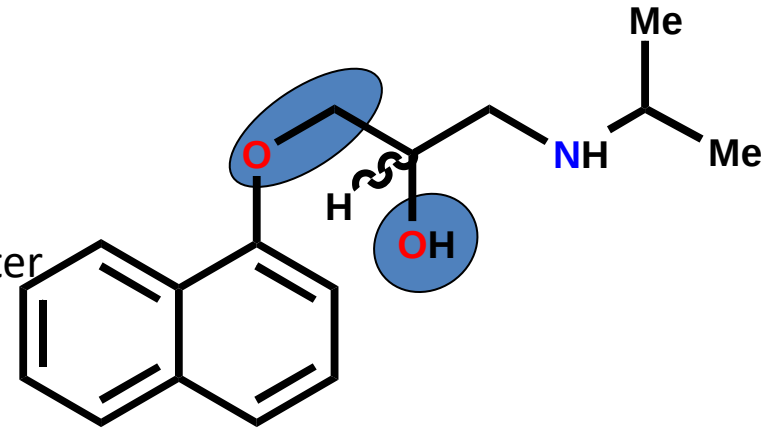
polarity, electronic distribution, and bonding

Replacing OH with the larger SH …… the electronic character

steric factors

Replacing OH with CH_3……. eliminate hydrogen bonding

Replacing OH with NH_2



Propranolol (β-blocker)

3- The presence of fluorine in place of an enzymatically labile hydrogen can also disrupt an enzymatic reaction, as C–F bonds are not easily broken.

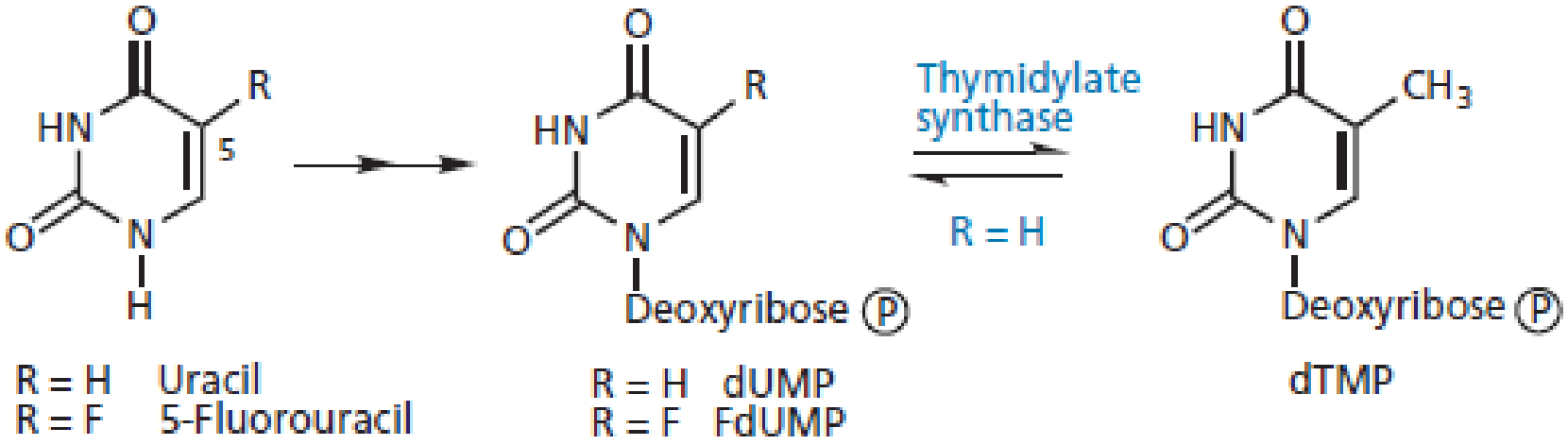


FIGURE 21.19 Biosynthesis of dTMP. Ⓟ = phosphate.

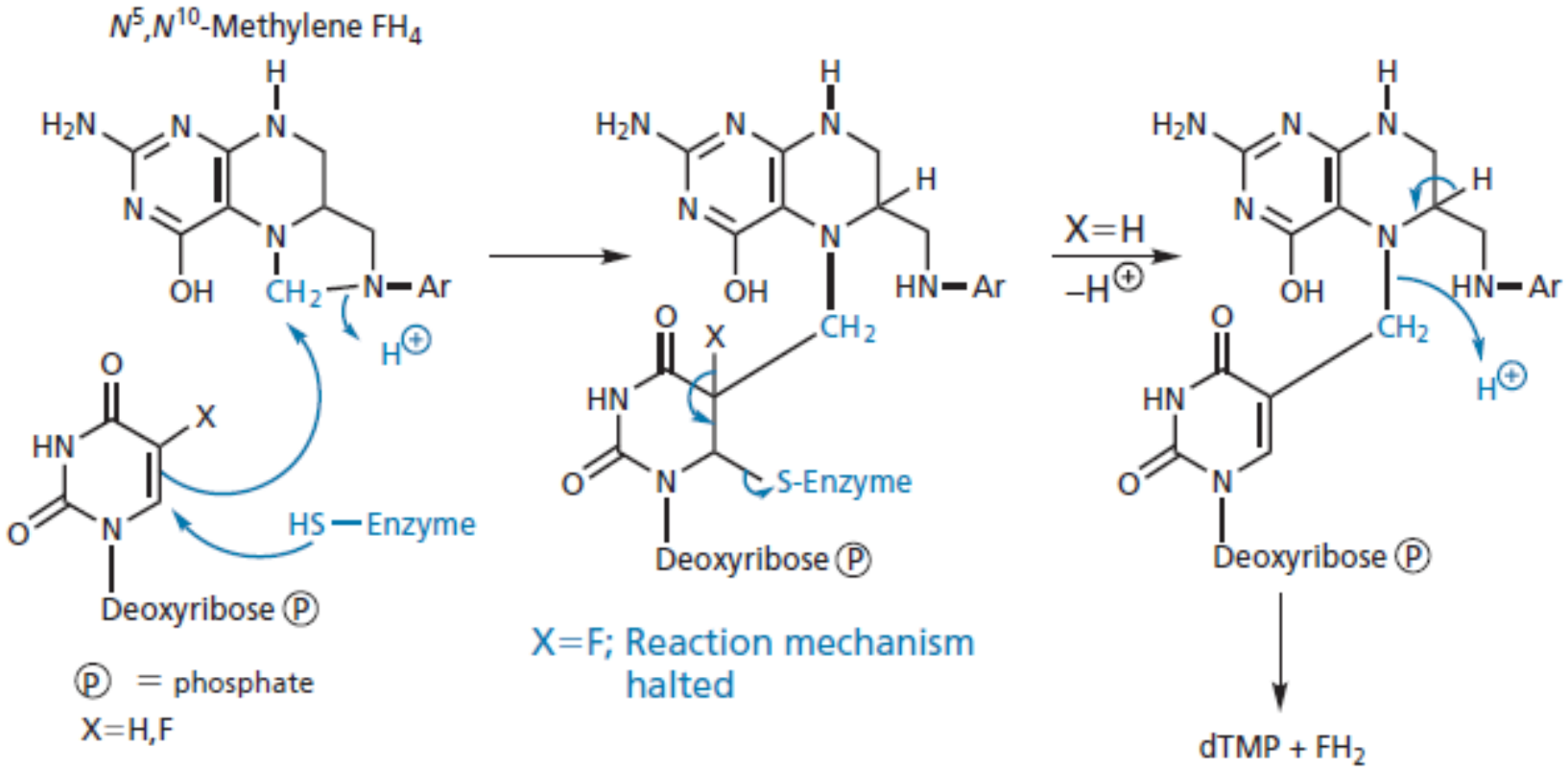
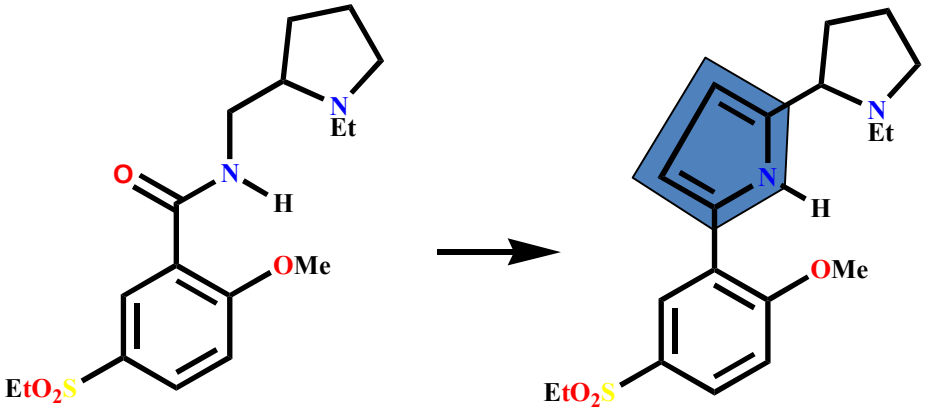


FIGURE 21.20 Use of 5-fluorouracil as a prodrug for a suicide substrate.

4- Improve activity and selectivity

- Replace a functional group with another group which retains the same biological activity
- Not necessarily the same valency

antipsychotics



Sultopride

dopamine antagonist

DU 122290

Improved selectivity for D_3 receptor over D_2 receptor

Pyrrole ring = bio-isostere for amide group

8-Simplification

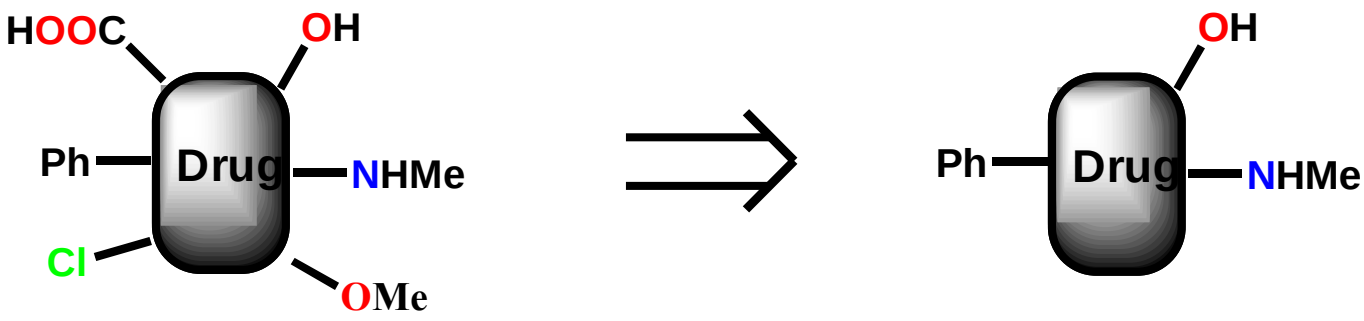
Rationale :

- Lead compounds from natural sources are often complex and difficult to synthesise.
- Simplifying the molecule makes synthesis of analogues easier, quicker and cheaper.
- Simpler structures may fit binding site easier and increase activity.
- Simpler structures may be more selective and less toxic if excess functional groups removed

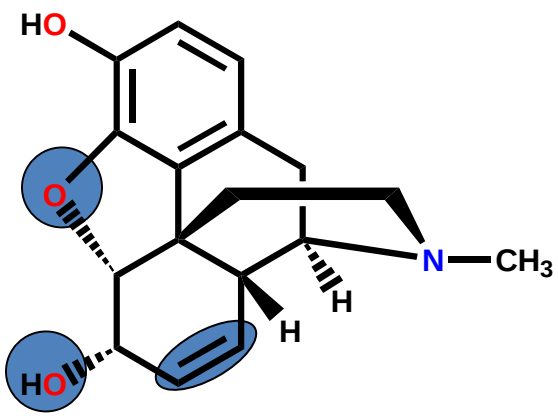
Methods

Retain pharmacophore

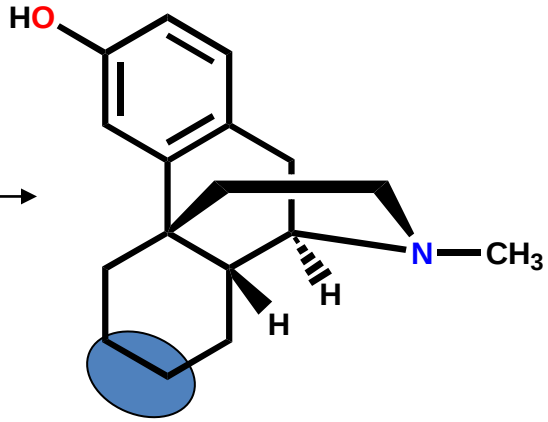
Remove unnecessary functional groups



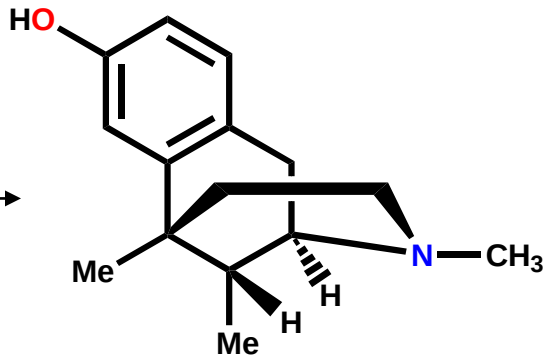
1- Remove excess rings



Morphine



Levorphanol



Metazocine

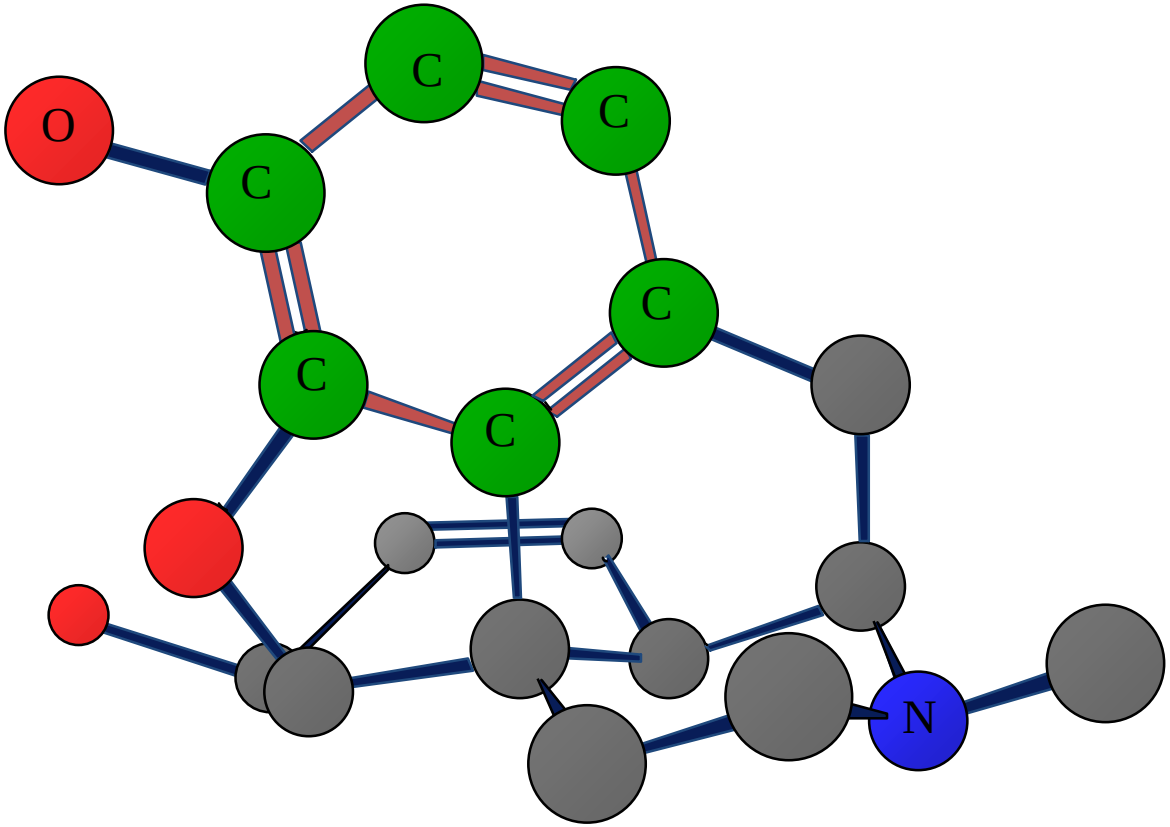
5X more potent than morphine

same potency as morphine

Excess functional groups

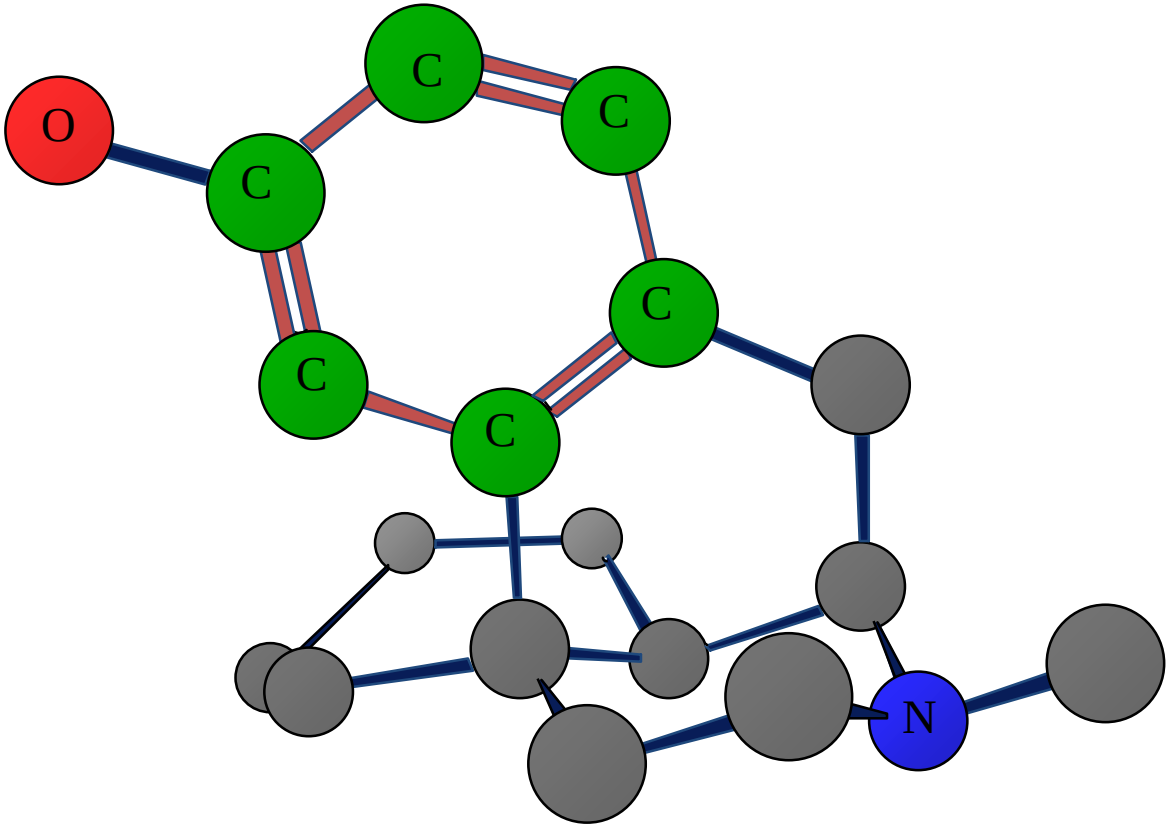
Excess ring

MORPHINE



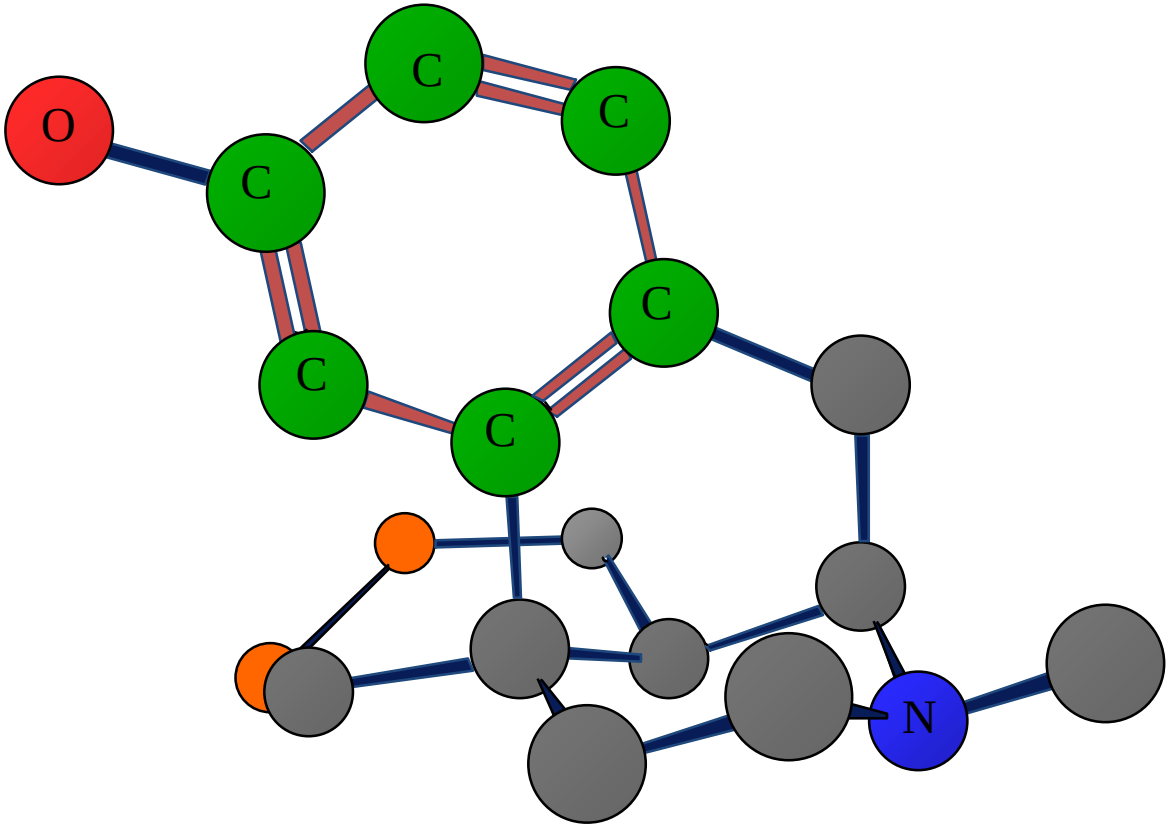
SIMPLIFICATION

LEVORPHANOL



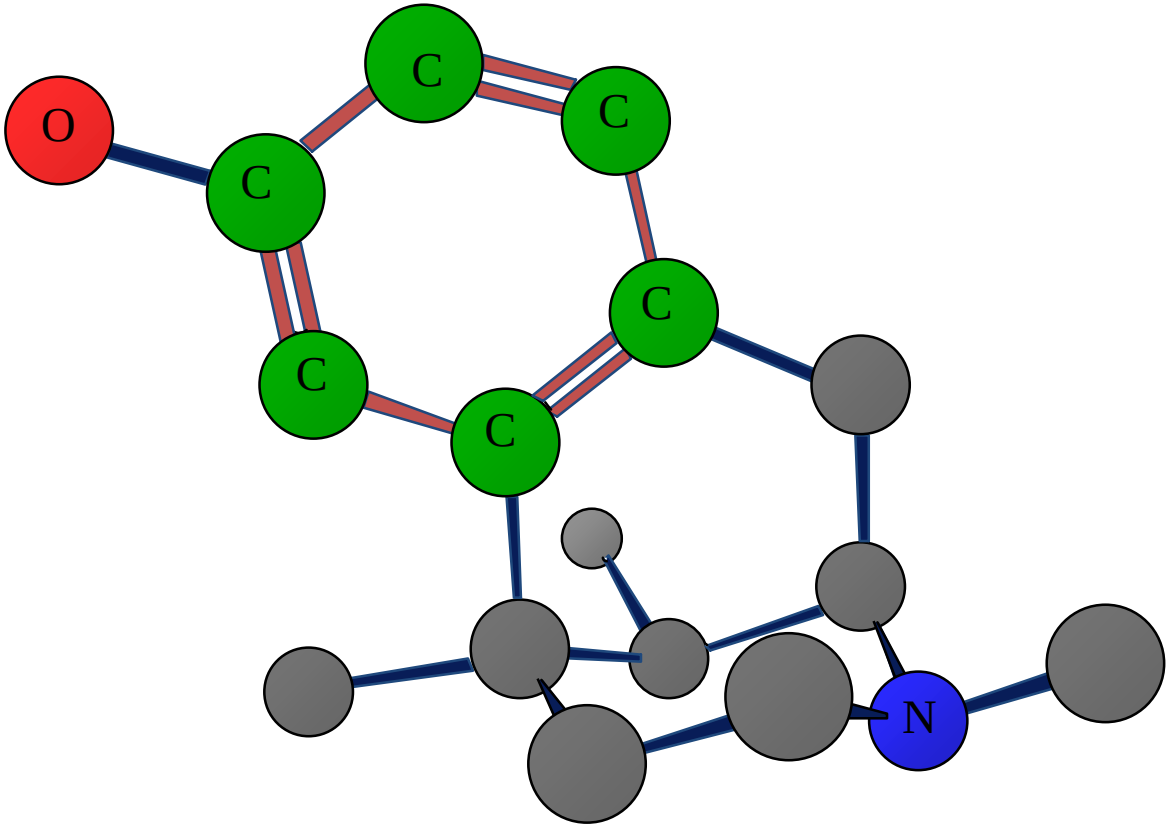
SIMPLIFICATION

LEVORPHANOL

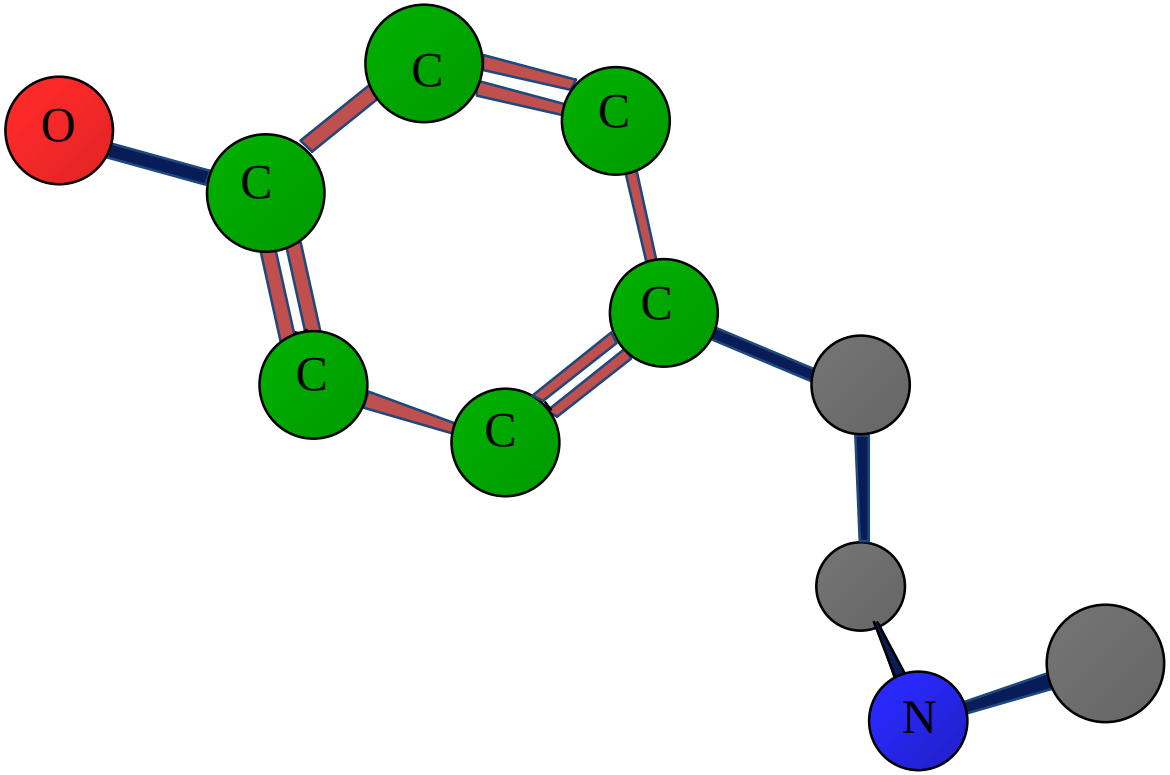


SIMPLIFICATION

METAZOCINE

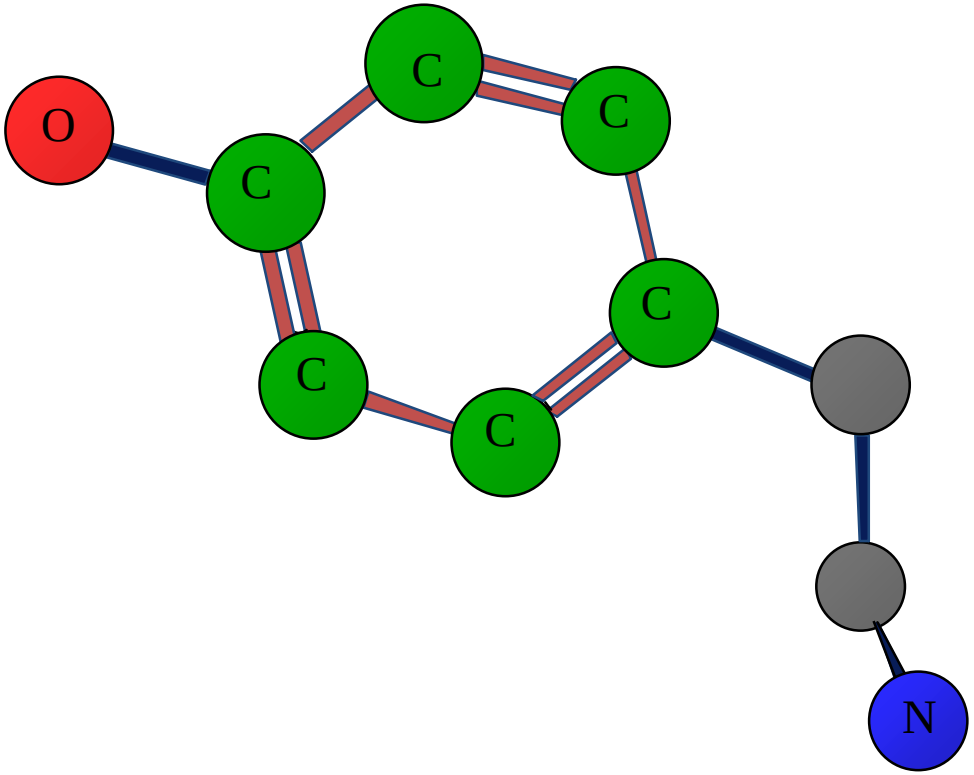


SIMPLIFICATION



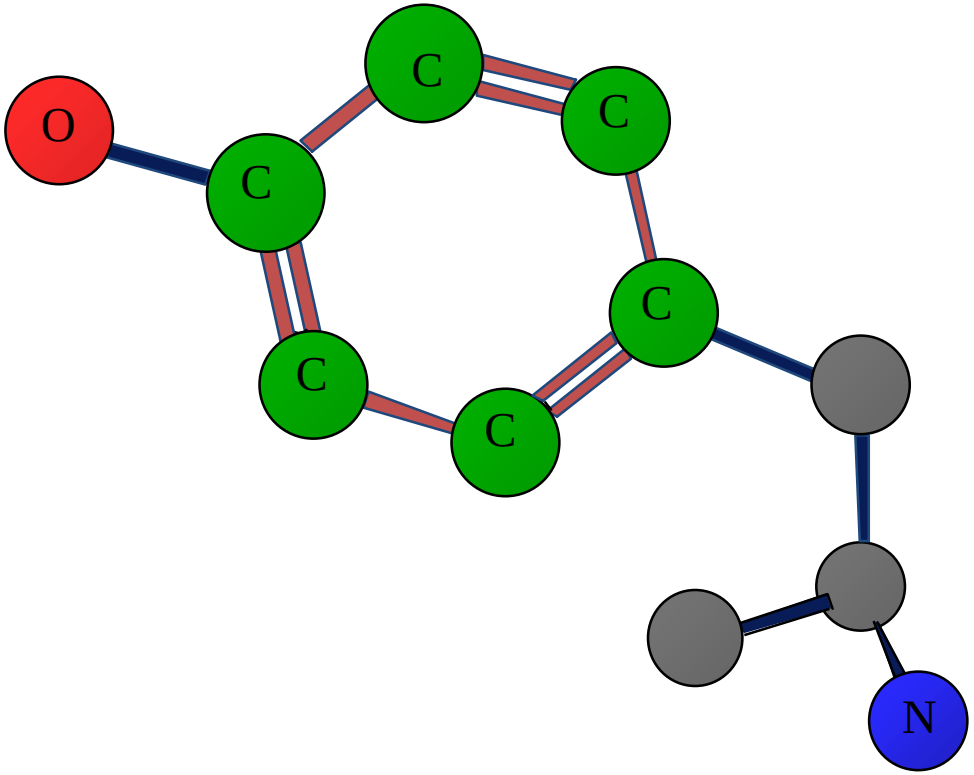
OVERSIMPLIFICATION

TYRAMINE

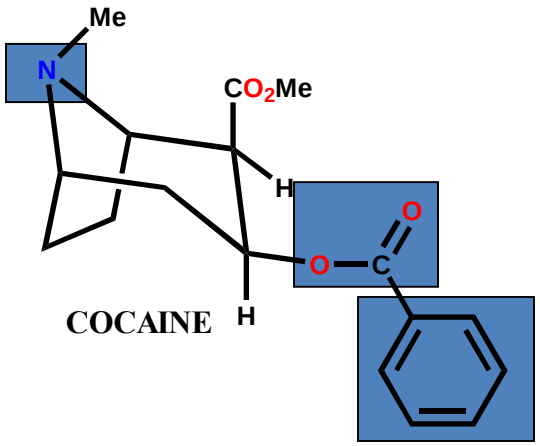


OVERSIMPLIFICATION

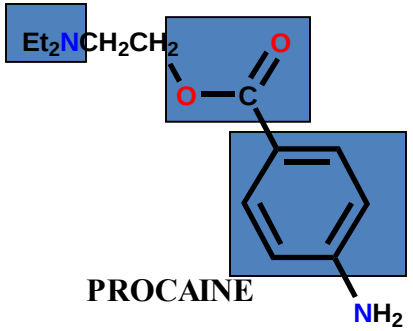
AMPHETAMINE



OVERSIMPLIFICATION



COCAINE

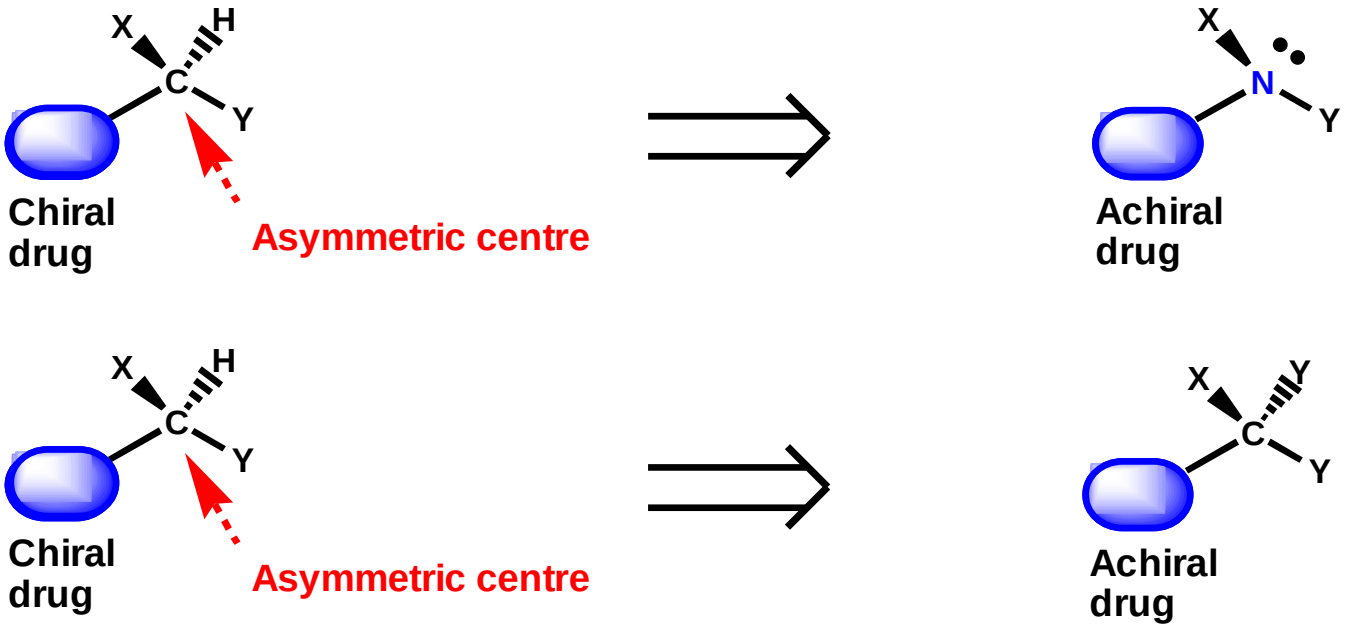


PROCAINE

local anaesthetics

Pharmacophore

2-Remove asymmetric centres



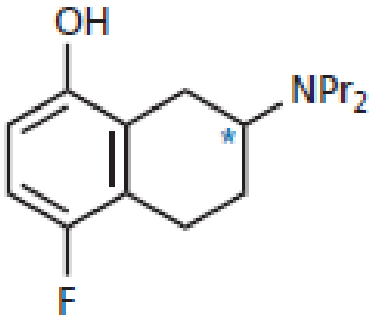
Why the use of racemates is discouraged and it is preferable to use a pure enantiomer ?

1-High cost of synthesis and seperation.

2-Both enantiomers then have to be tested for their activity and side effects, doubling the number of tests that have to be carried out.

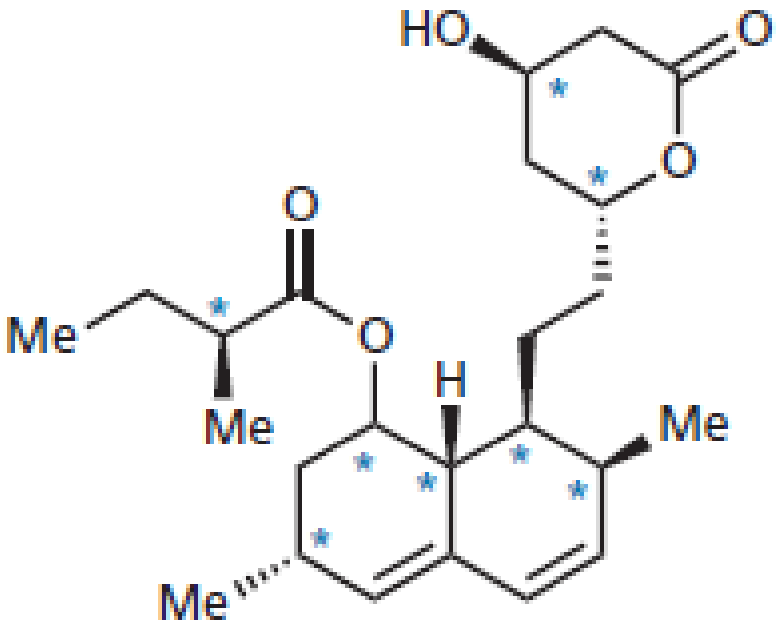
3-different enantiomers can have different activities.

Enantiomers have opposing agonist and antagonist activity at the serotonin receptor (5-HT 1A)

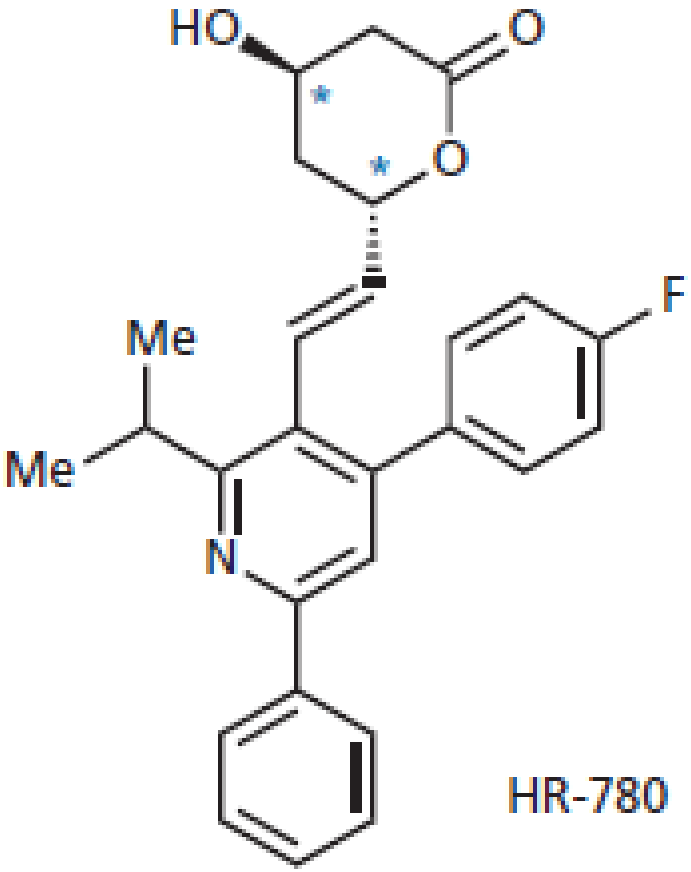


UH-301

Inactive as a racemate



Mevinolin (lovastatin)



HR-780

cholesterol-lowering agents

Disadvantages of simplification:

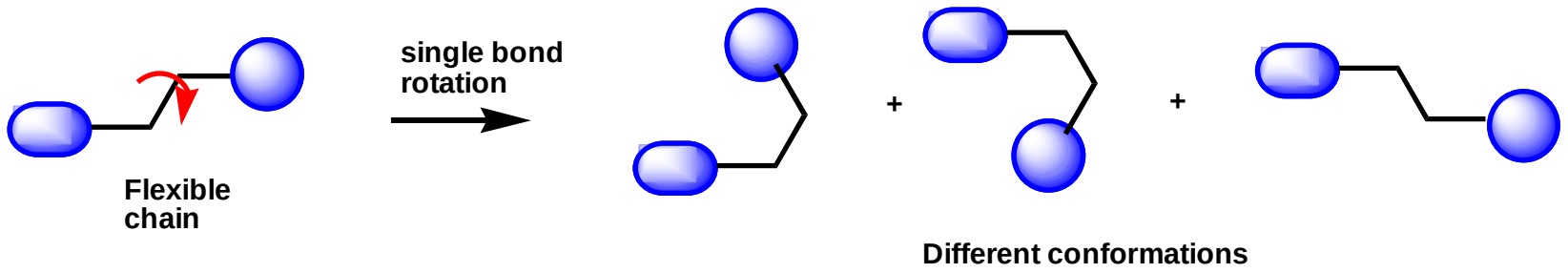
- Oversimplification may result in decreased activity and selectivity. e.g morphine over simplication
- Simpler molecules have more conformations bind in a different way than the lead compound.
- More likely to interact with more than one target binding site more side effects.

9- Rigidification

Rationale :

Endogenous lead compounds often simple and flexible (e.g. adrenaline)

Fit several targets due to different active conformations (e.g. Adrenergic receptor types and subtypes)

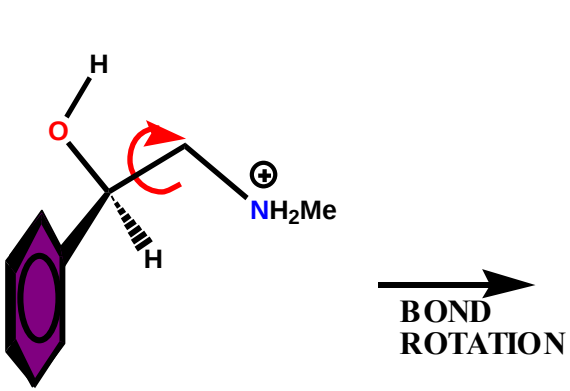


- Rigidify molecule to limit conformations - conformational restraint
- Increases activity (more chance of desired active conformation)
- Increases selectivity (less chance of undesired active conformations)

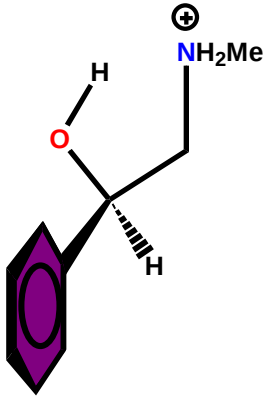
Disadvantage:

- Molecule more complex and may be more difficult to synthesise

The body's own neurotransmitters

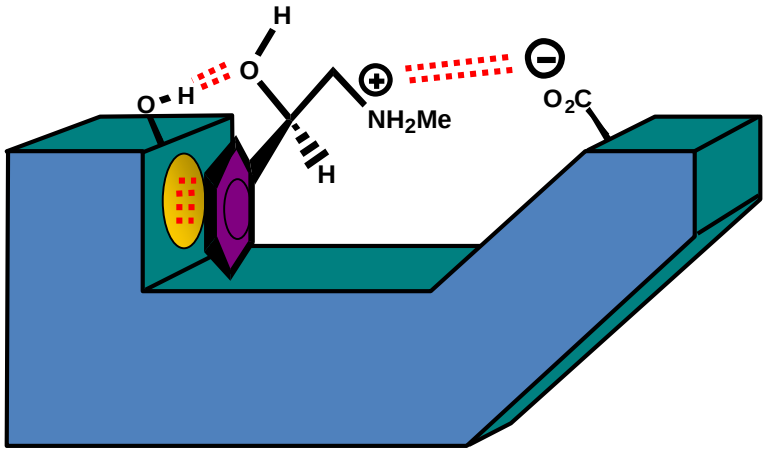


BOND ROTATION

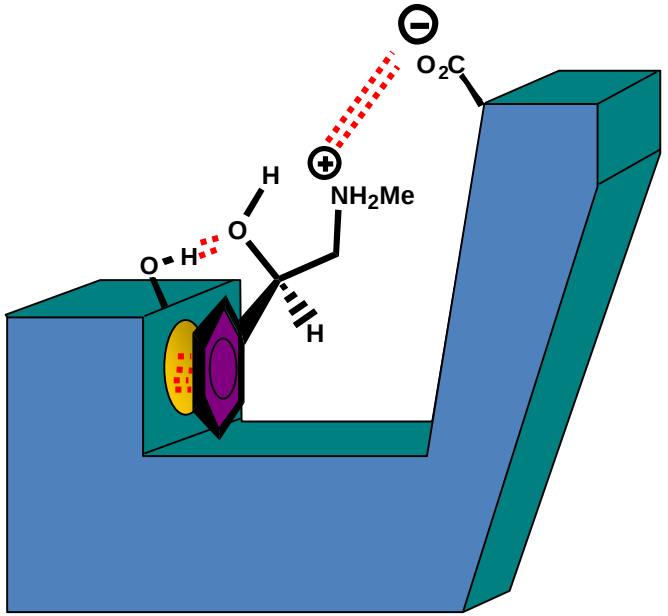


I

II



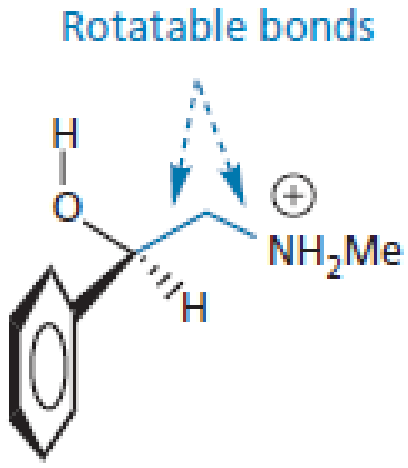
RECEPTOR 1



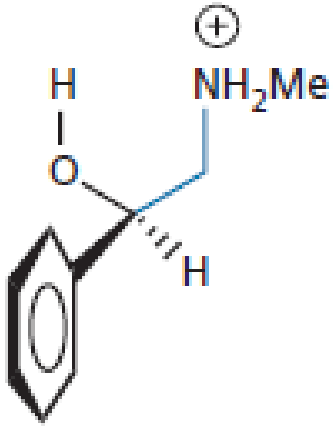
RECEPTOR 2

1- Introduce rings

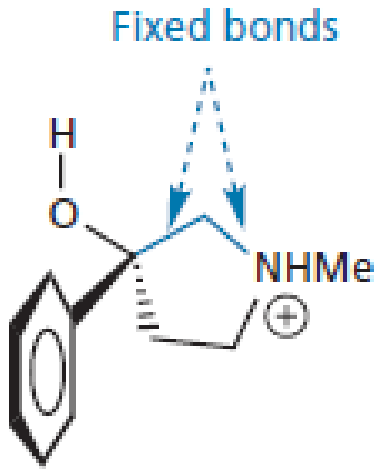
Bonds within ring systems are locked and cannot rotate freely



Flexible messenger
in the active conformation

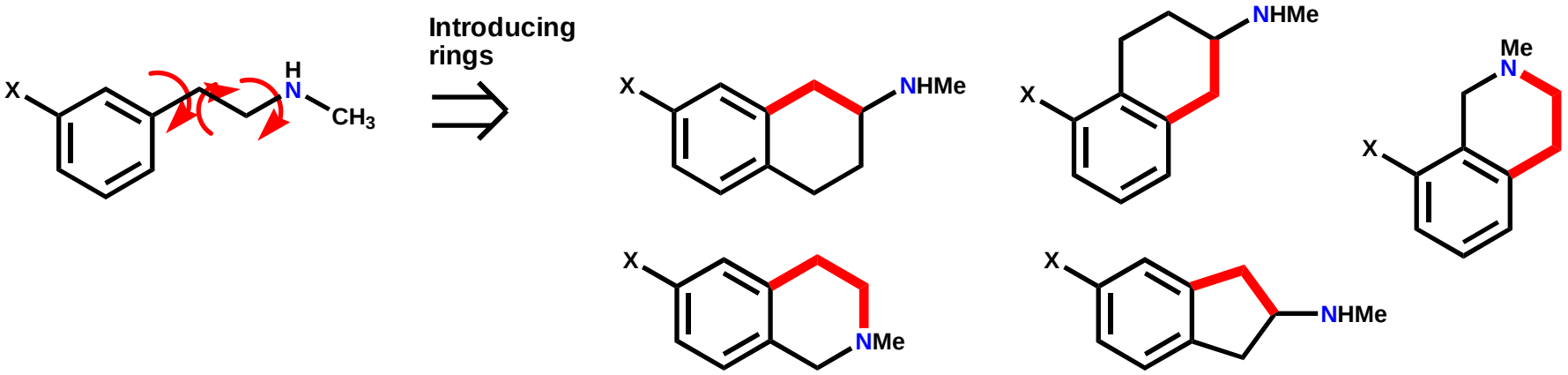


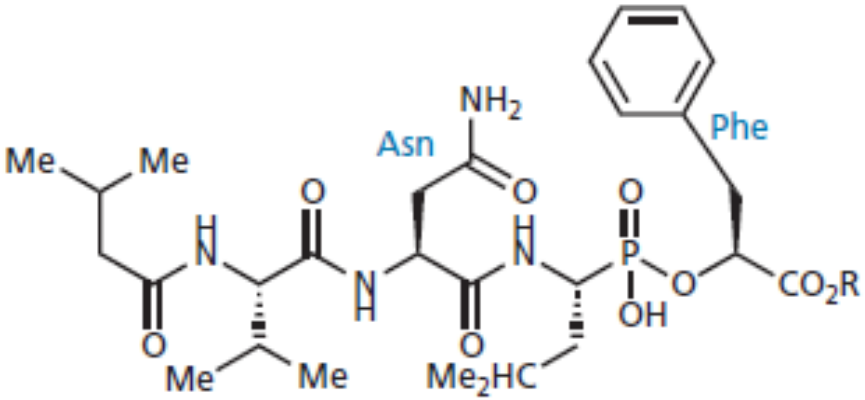
Flexible messenger
in an inactive conformation



Rigid messenger held
in the active conformation

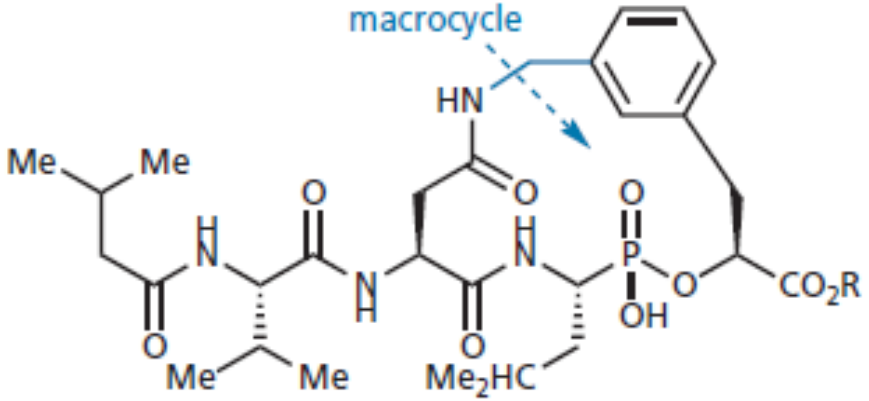
Test rigid structures to see which ones have retained active conformation





Acyclic pentapeptide (K_i 42 nM)

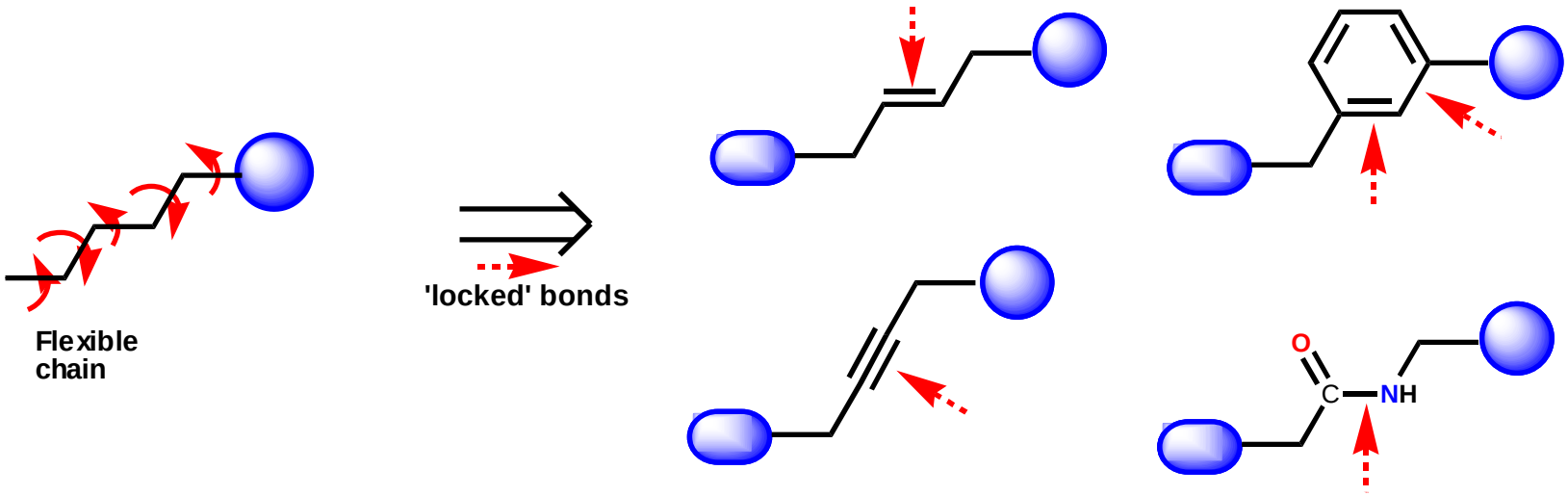
inhibitor of a proteolytic enzyme



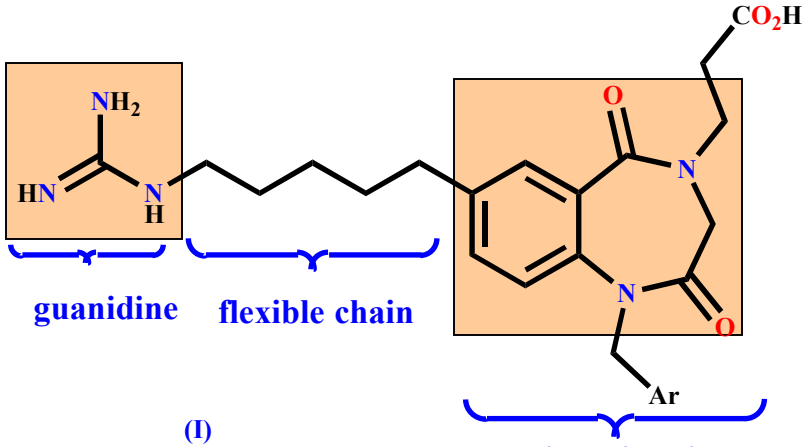
Rigidified pentapeptide (K_i 0.1 nM)

400-fold increase in activity

2-Introduce rigid functional groups



Important binding groups

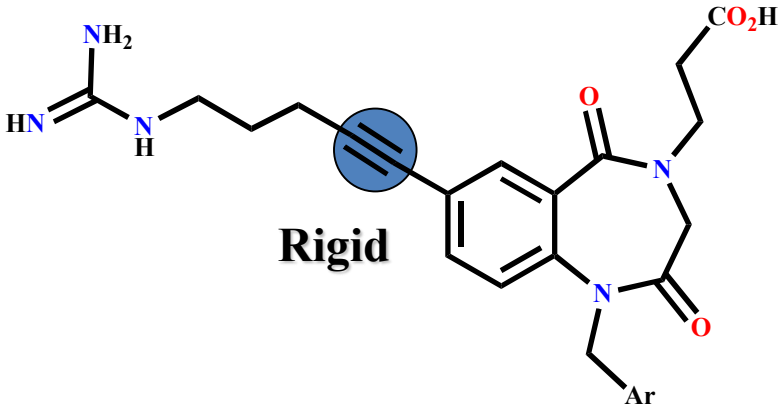


(I)

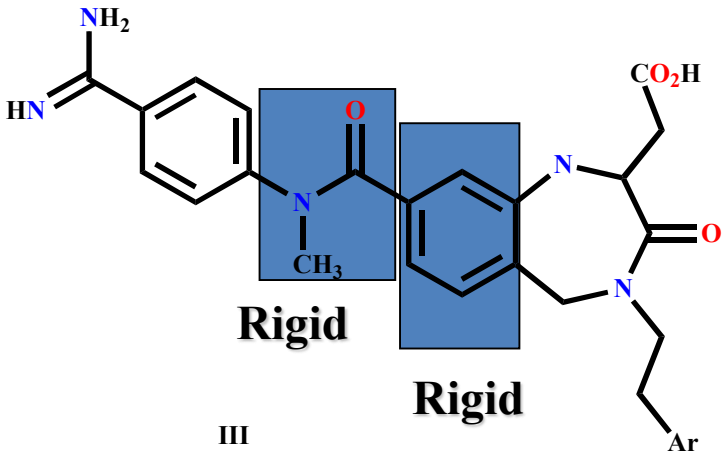
diazepin

Inhibits platelet aggregation

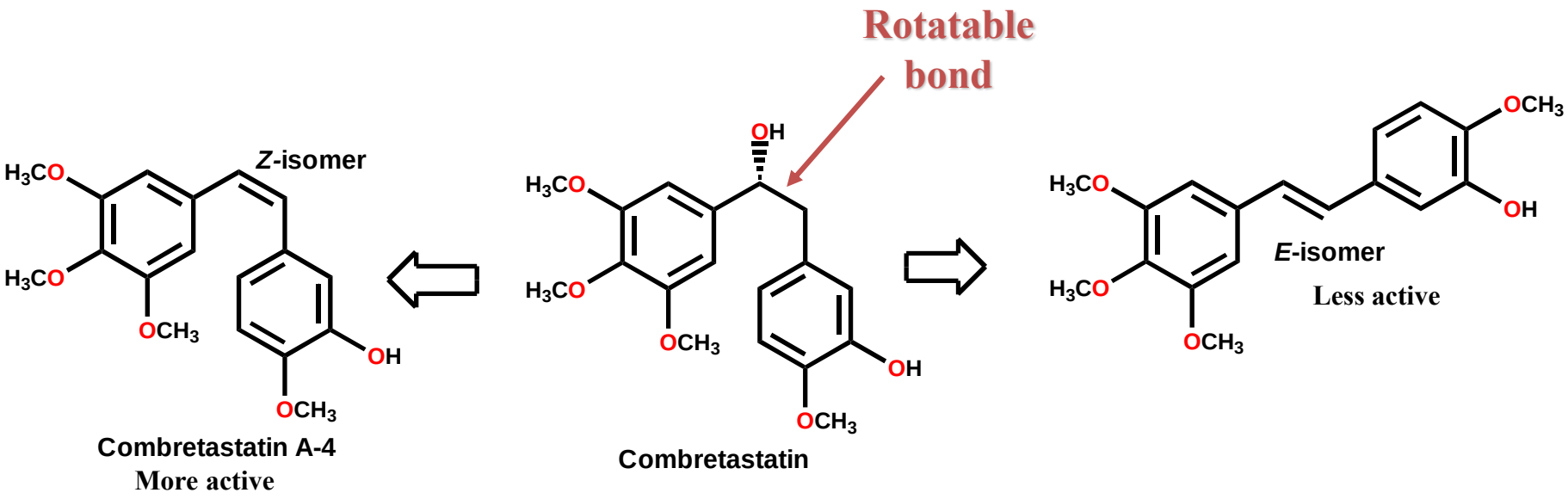
Analogues



II



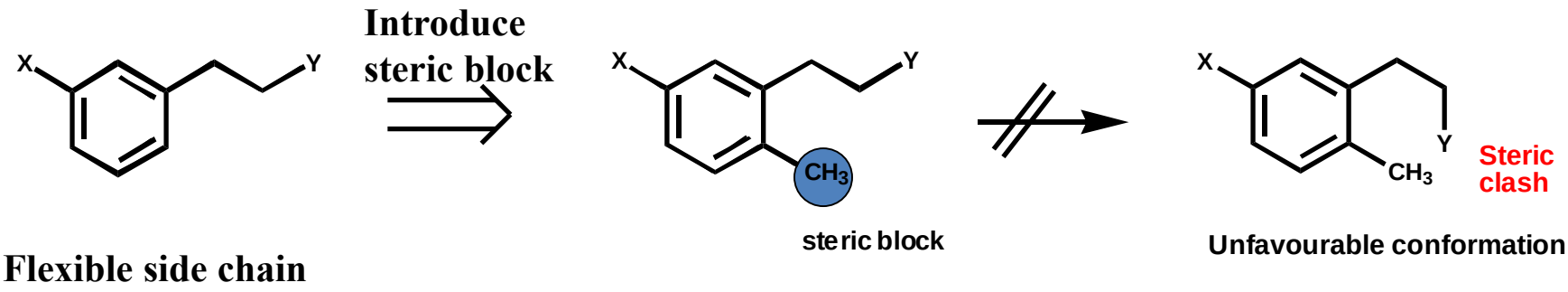
III

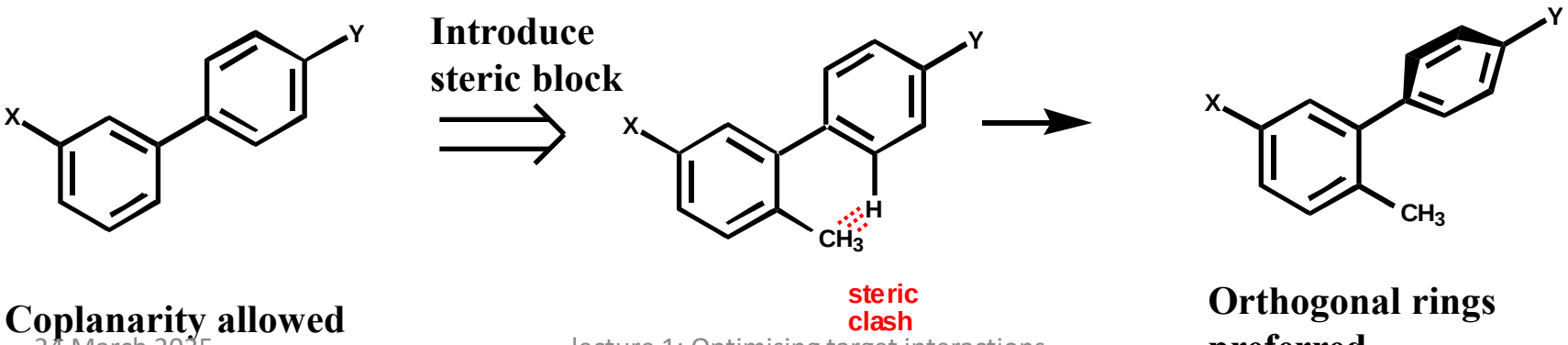


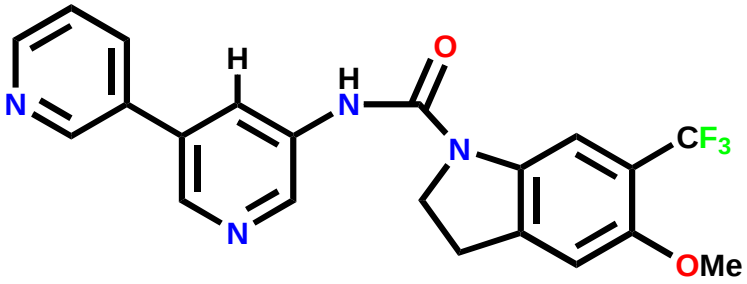
Combretastatin (anticancer agent)

3- Steric blockers

In certain situations, a quite simple substituent can hinder the free rotation of a single bond.



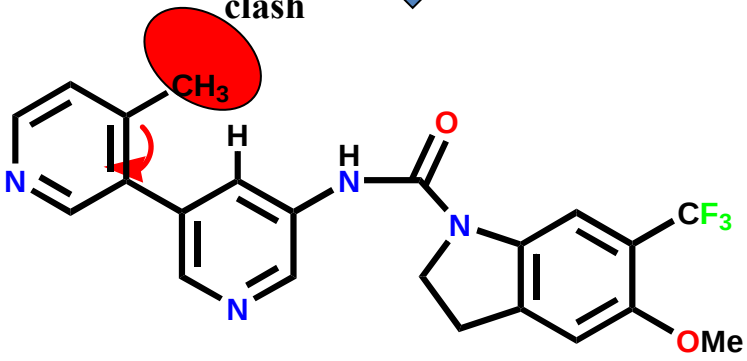


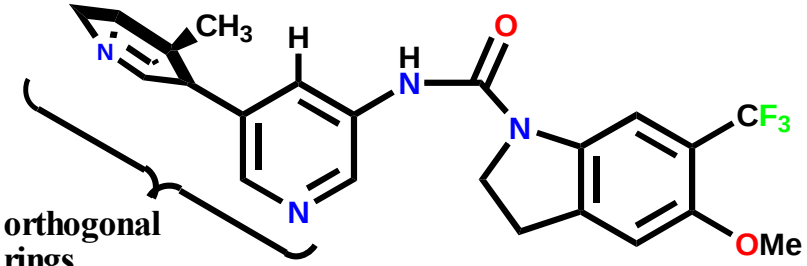


Serotonin antagonist

Introduce methyl group

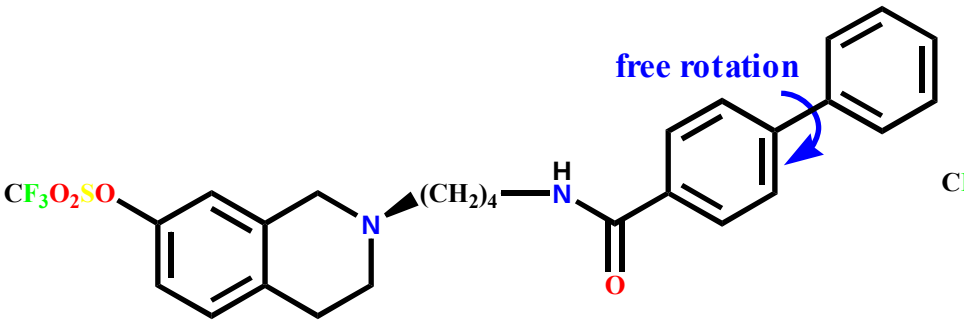
Steric clash





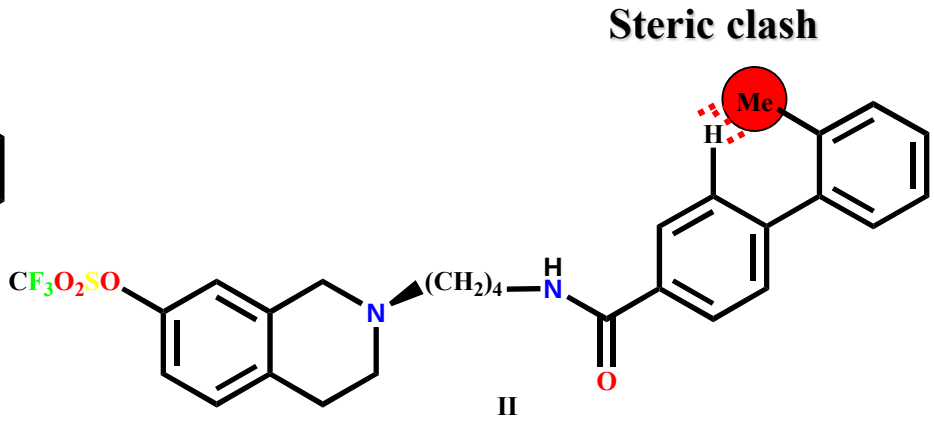
orthogonal rings

Increase in activity
Active conformation retained



I

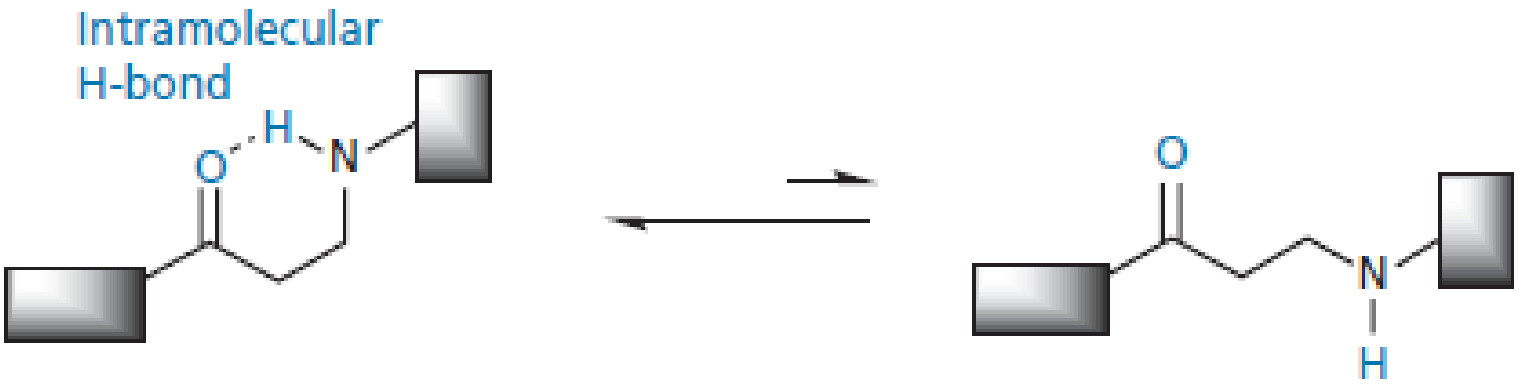
D_3 Antagonist



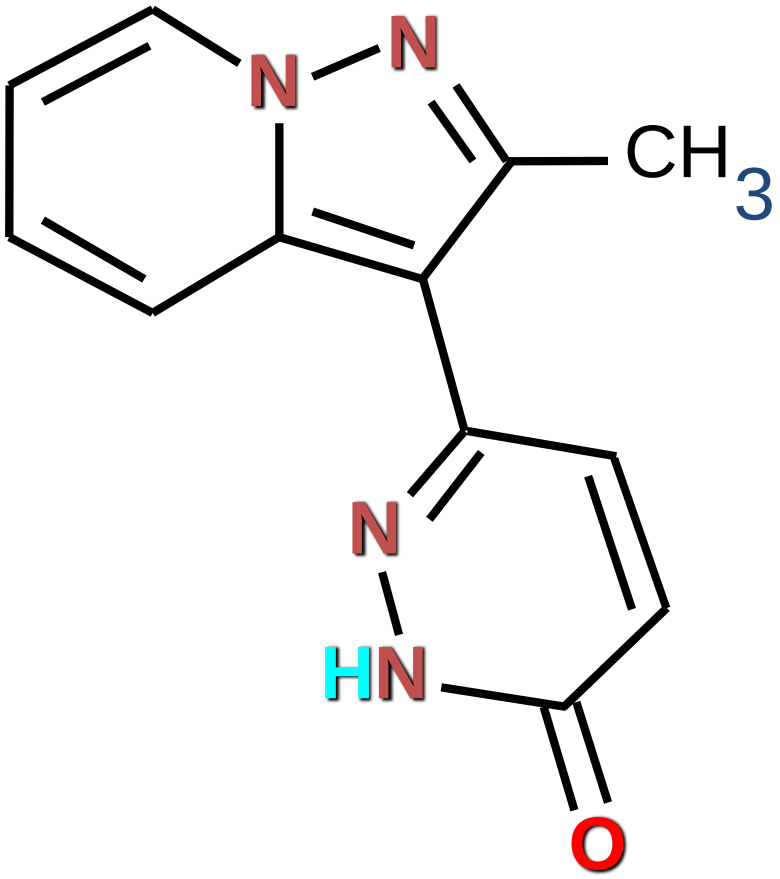
II

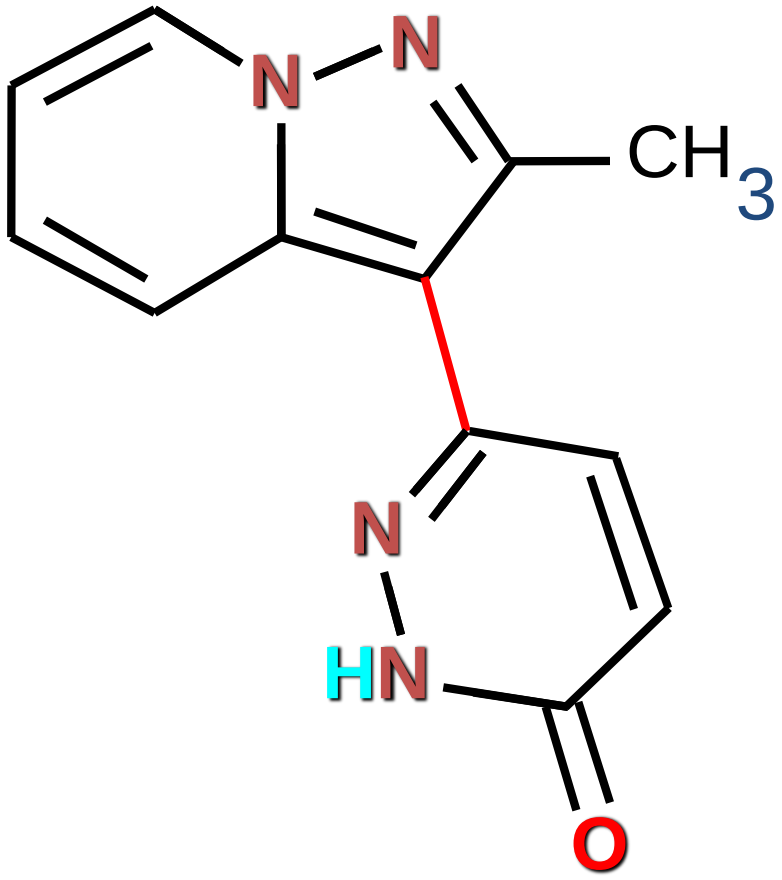
Inactive
Active conformation disallowed

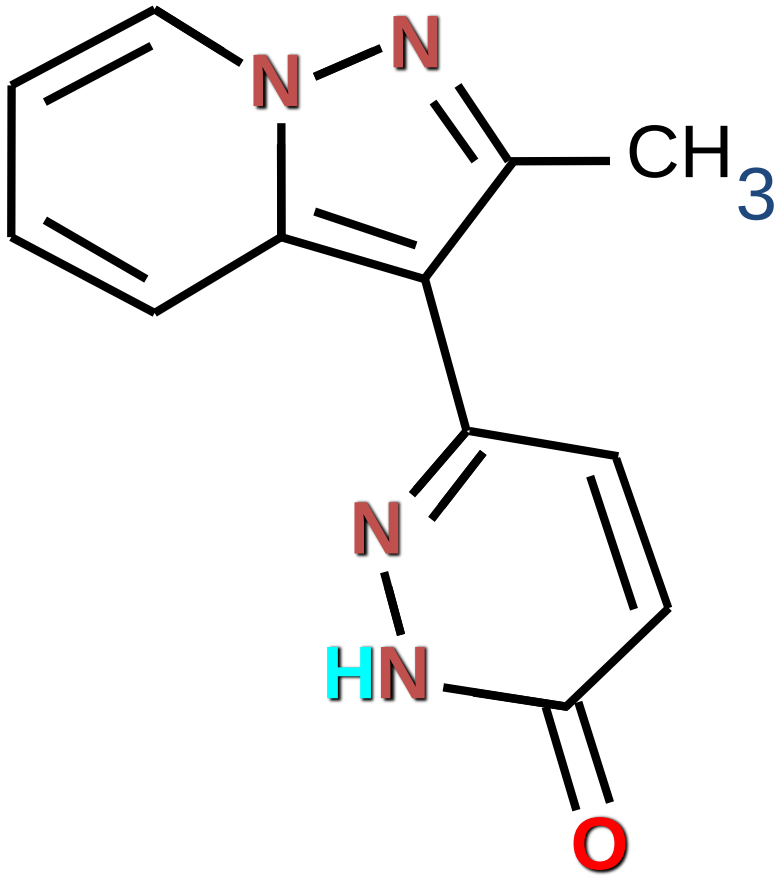
4-intramolecular hydrogen bonding

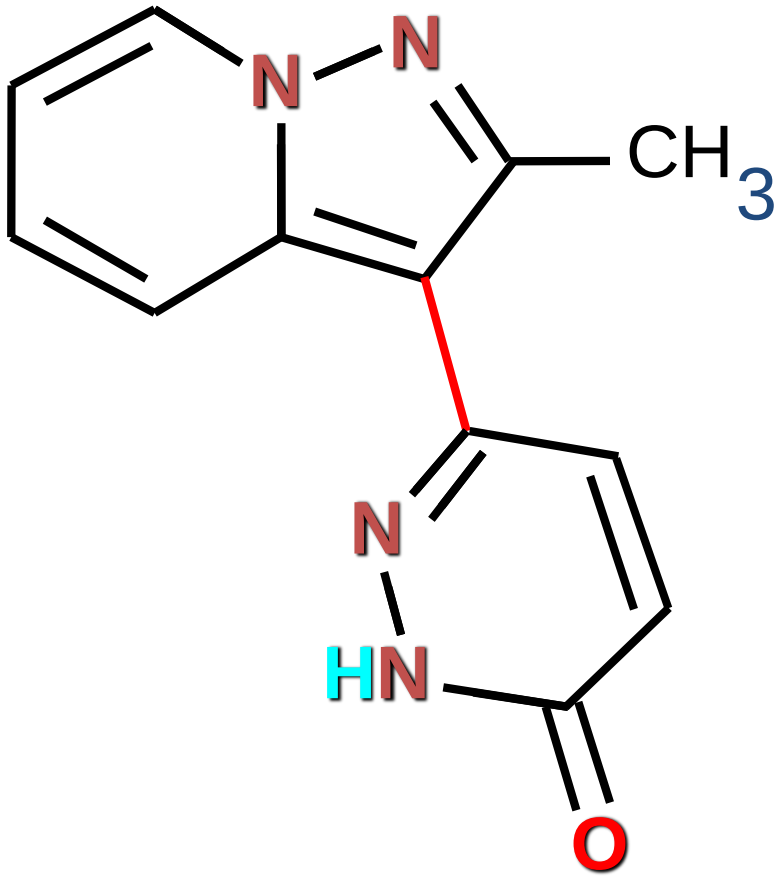


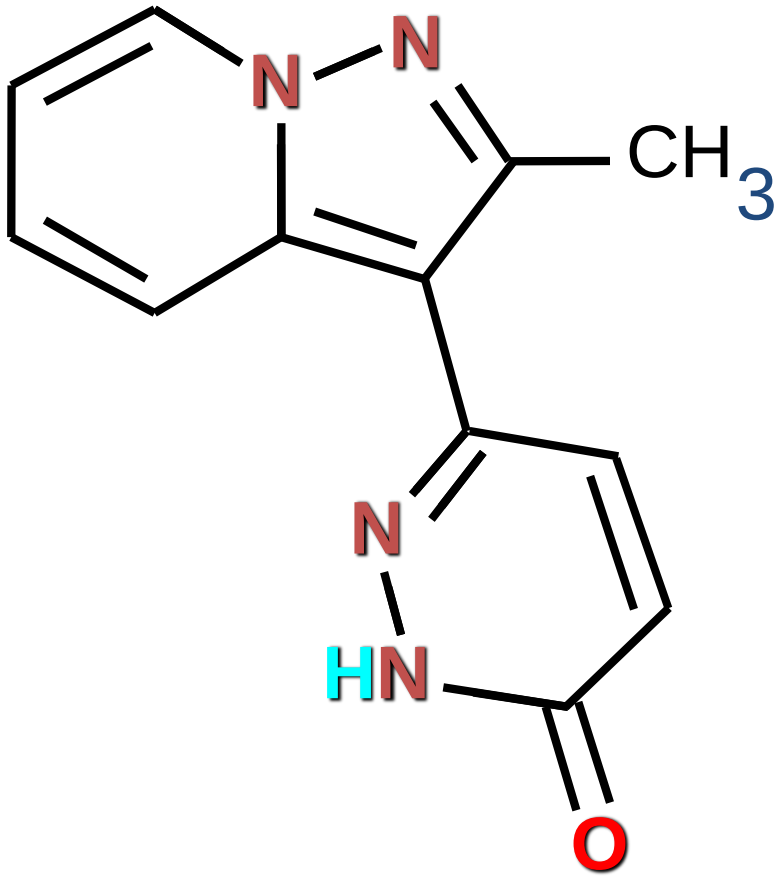
Identification of an active conformation by rigidification




N
N
CH3
N
HN
O

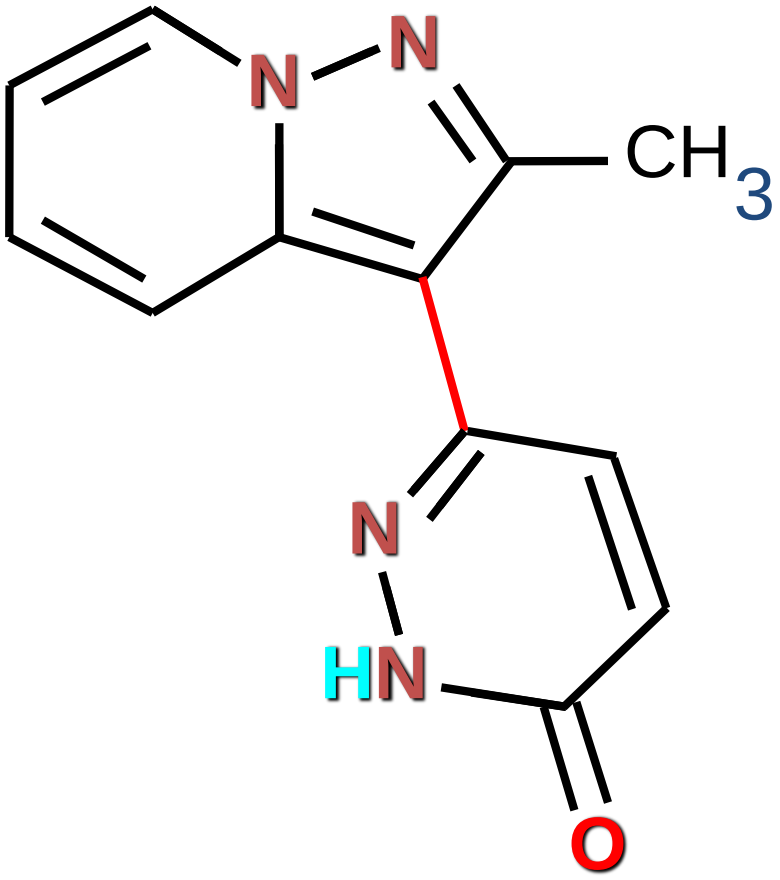

N
N
CH3
N
HN
O

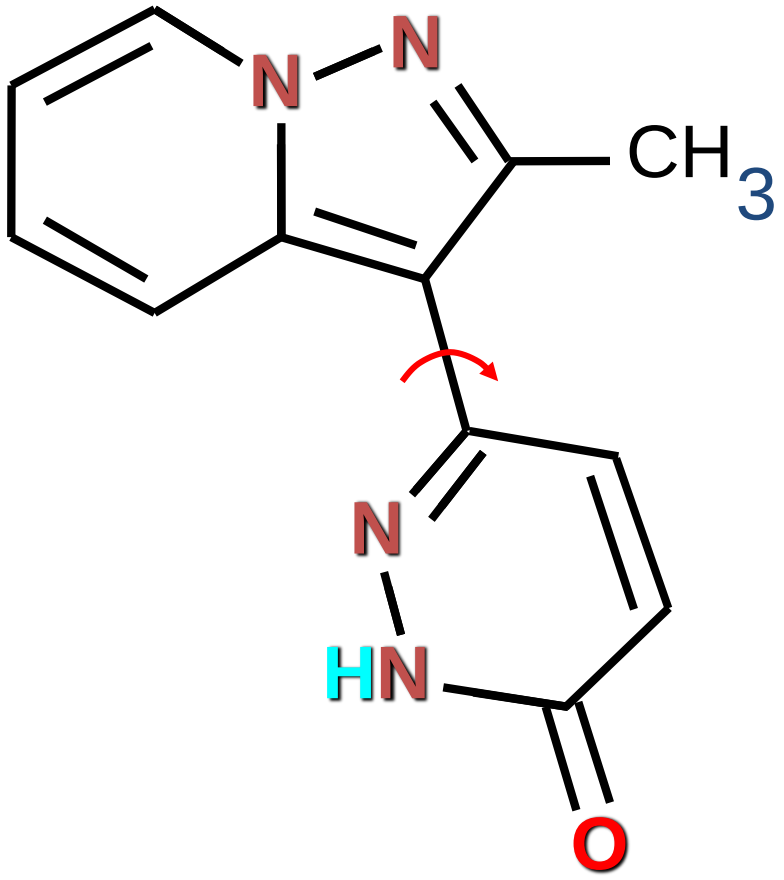

N
N
CH3
N
HN
O

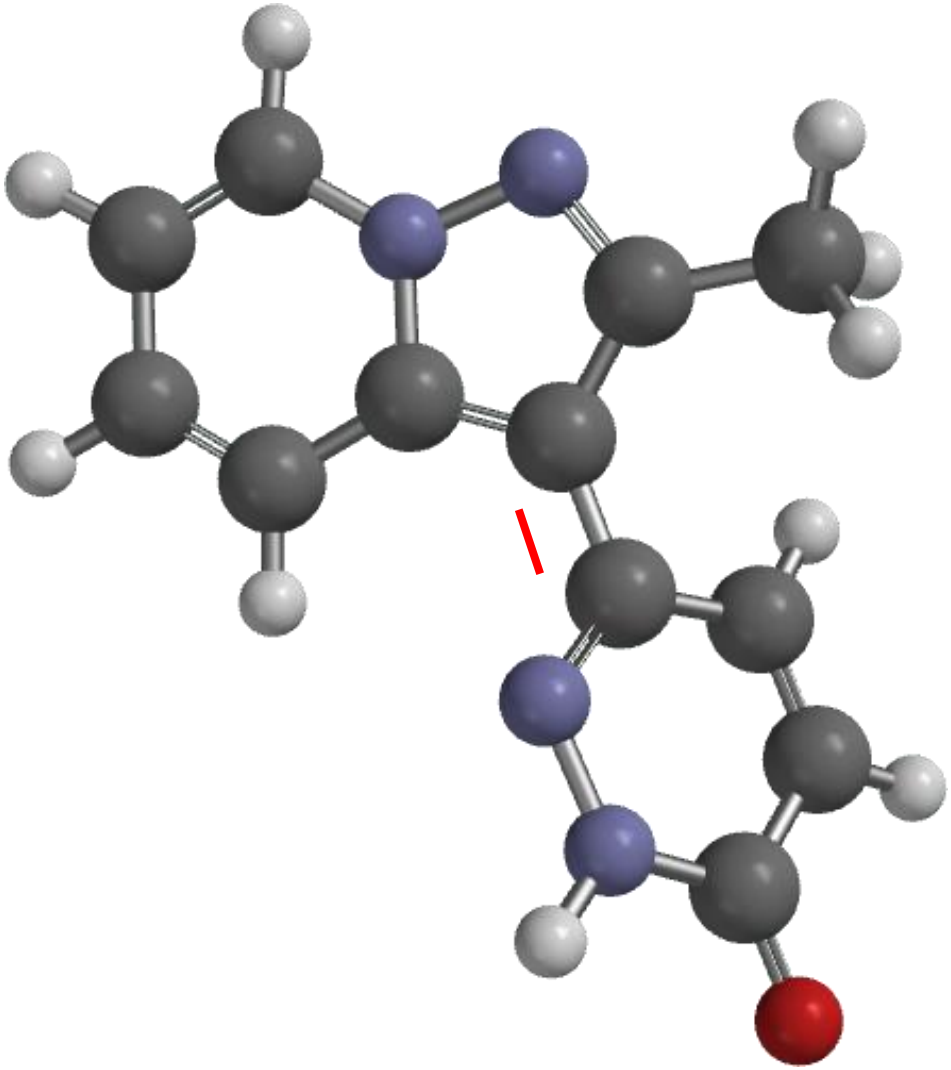

N
N
CH3
N
HN
O

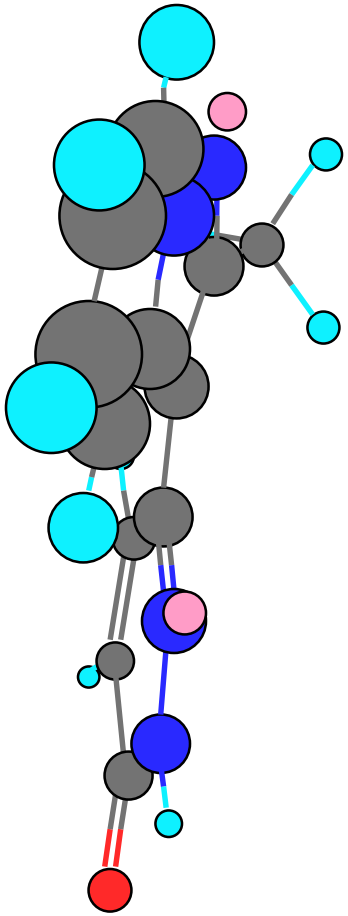
7.9 Rigidification

Identification of an active conformation by rigidification

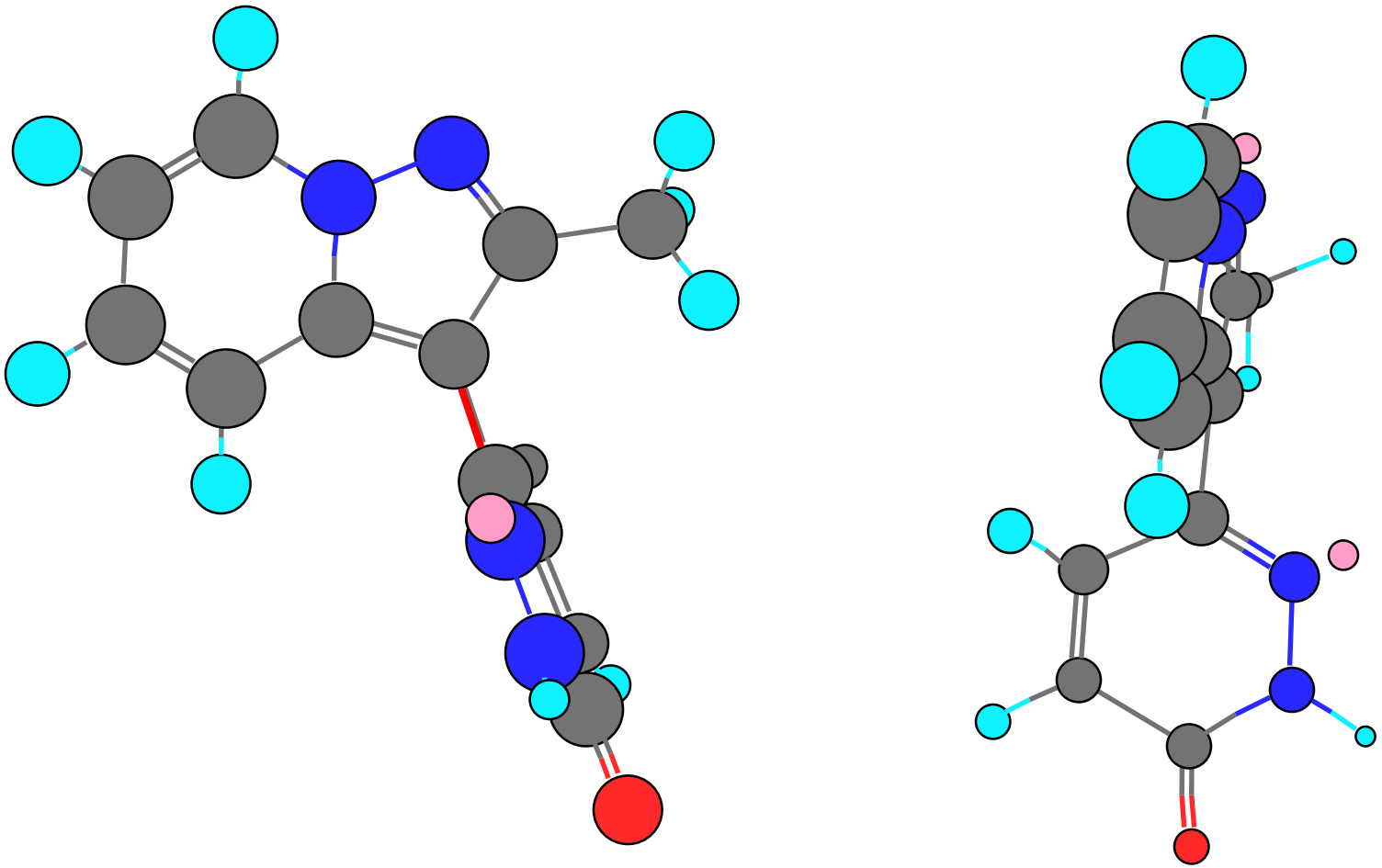



N
N
CH3
N
HN
O



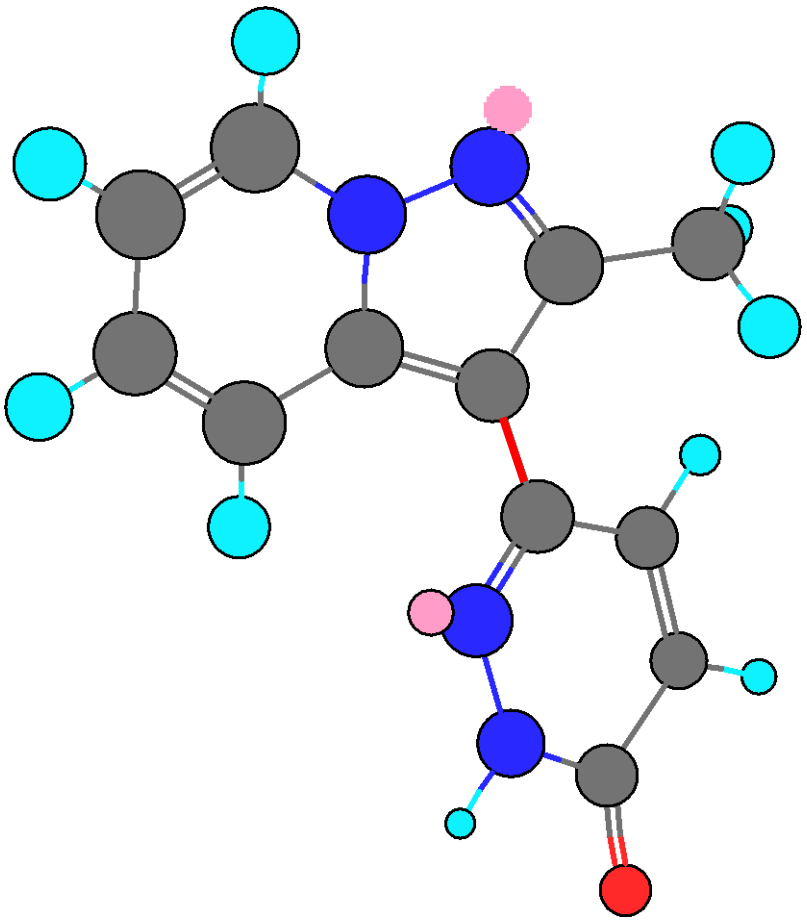


Planar conformation



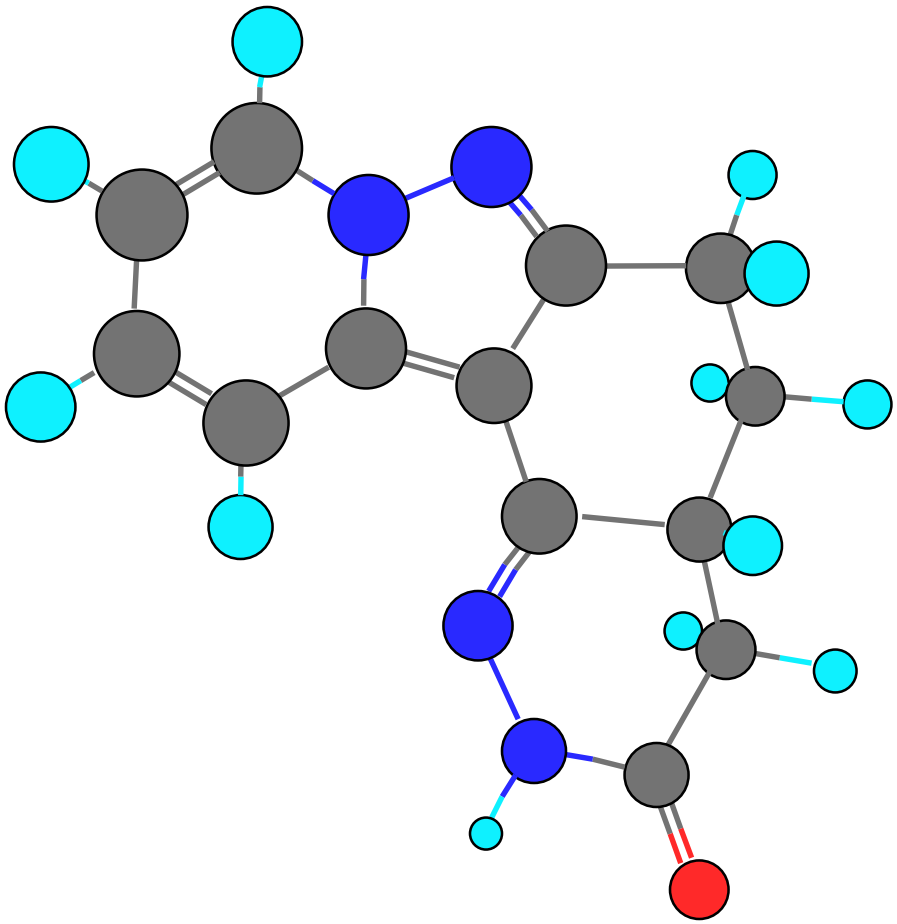
Orthogonal conformation

Redo 3d diagrams



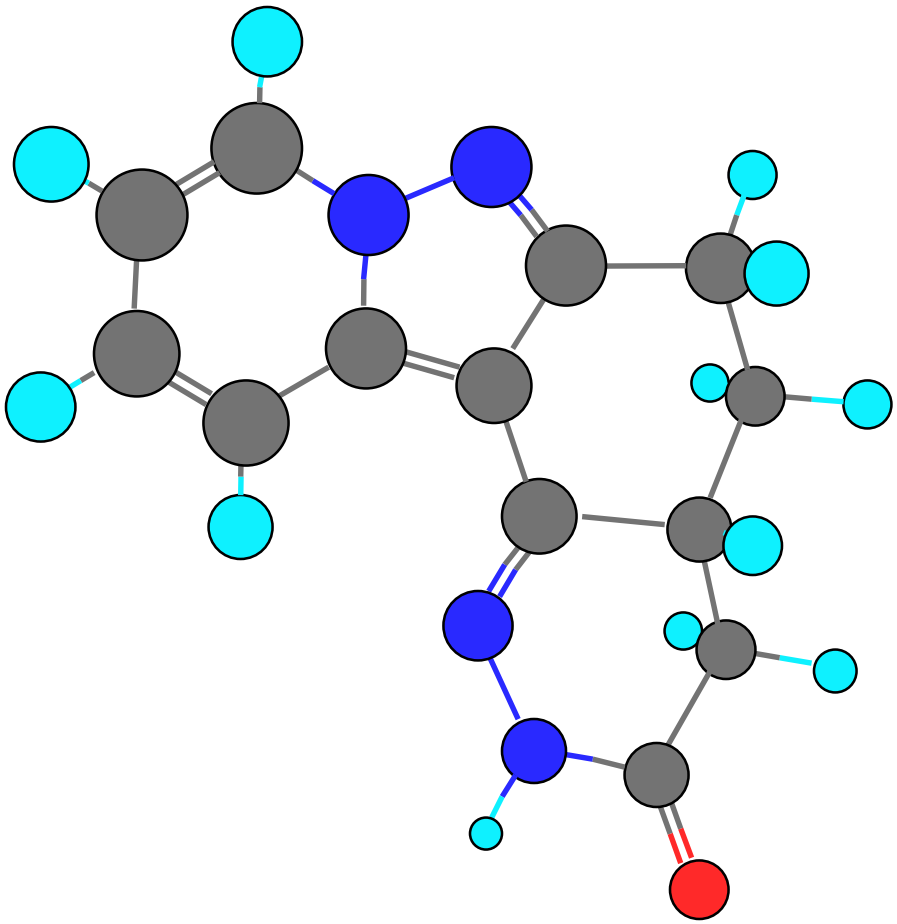
CYCLISATION

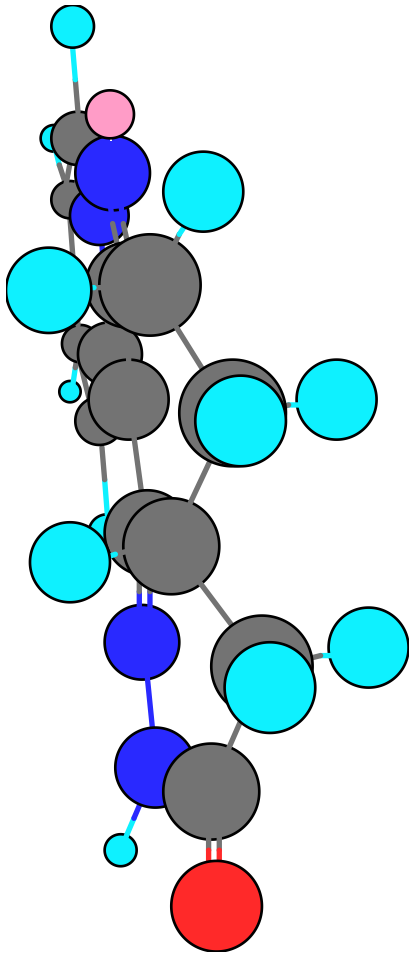
Redo 3d diagrams



CYCLISATION

Redo 3d diagrams

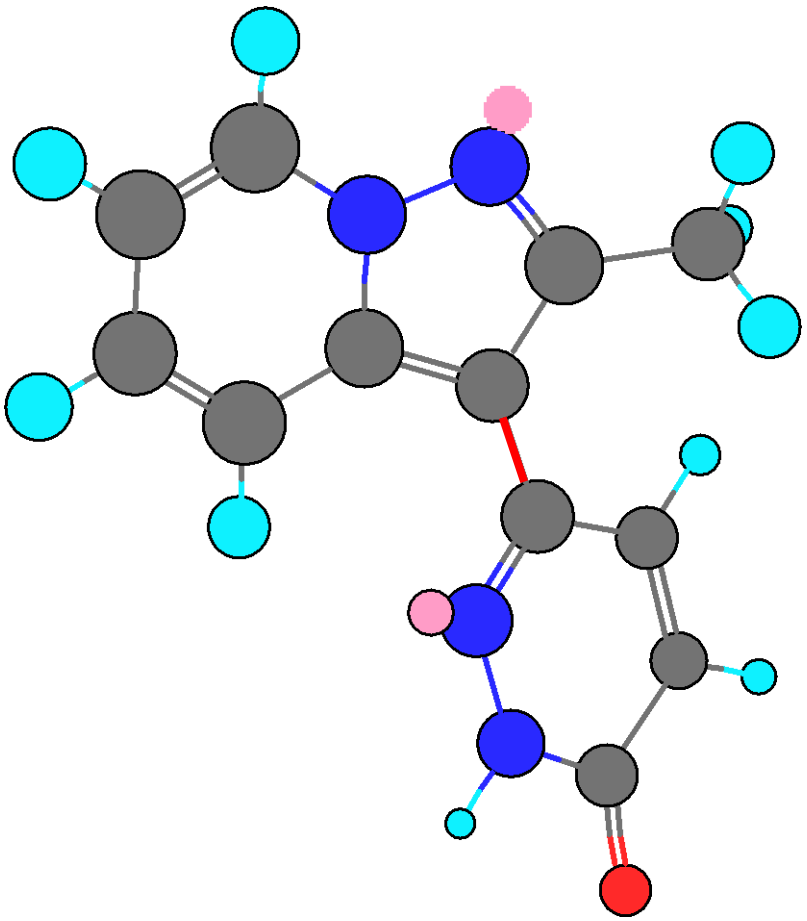




CYCLISATION

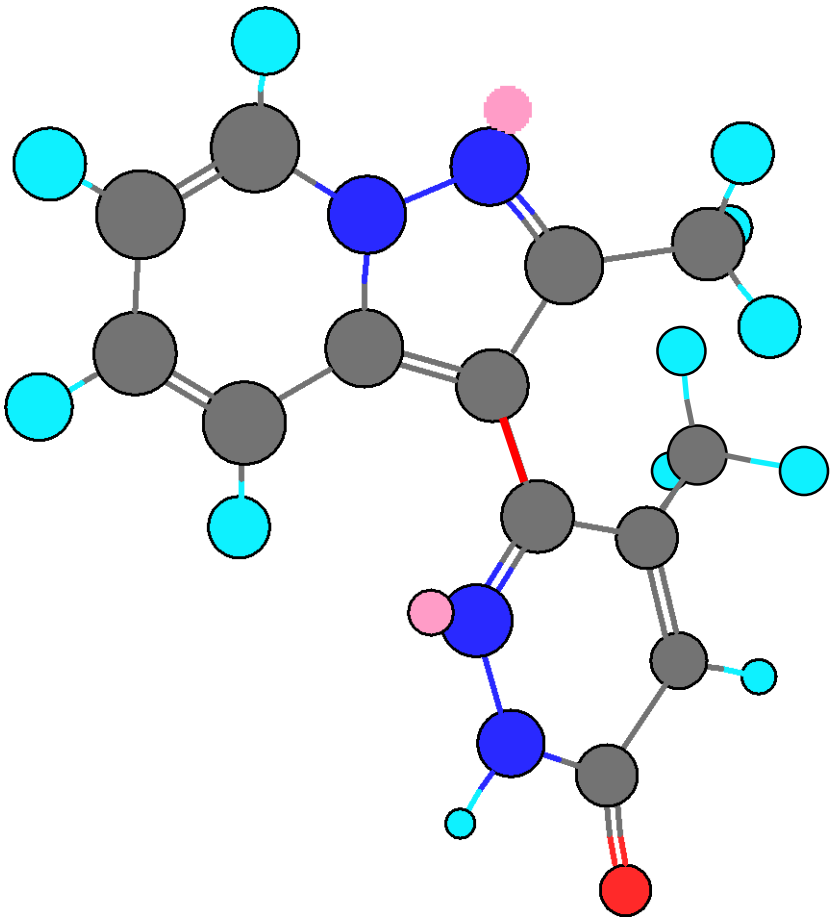
Locked into planar conformation

Redo 3d diagrams



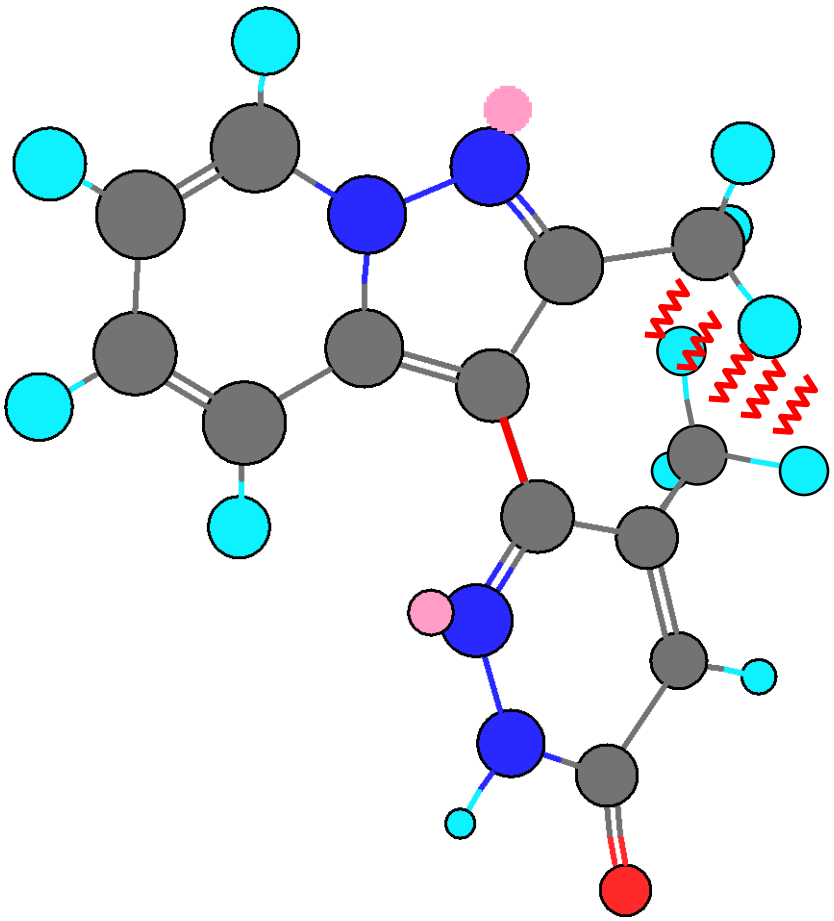
STERIC HINDRANCE

Redo 3d diagrams



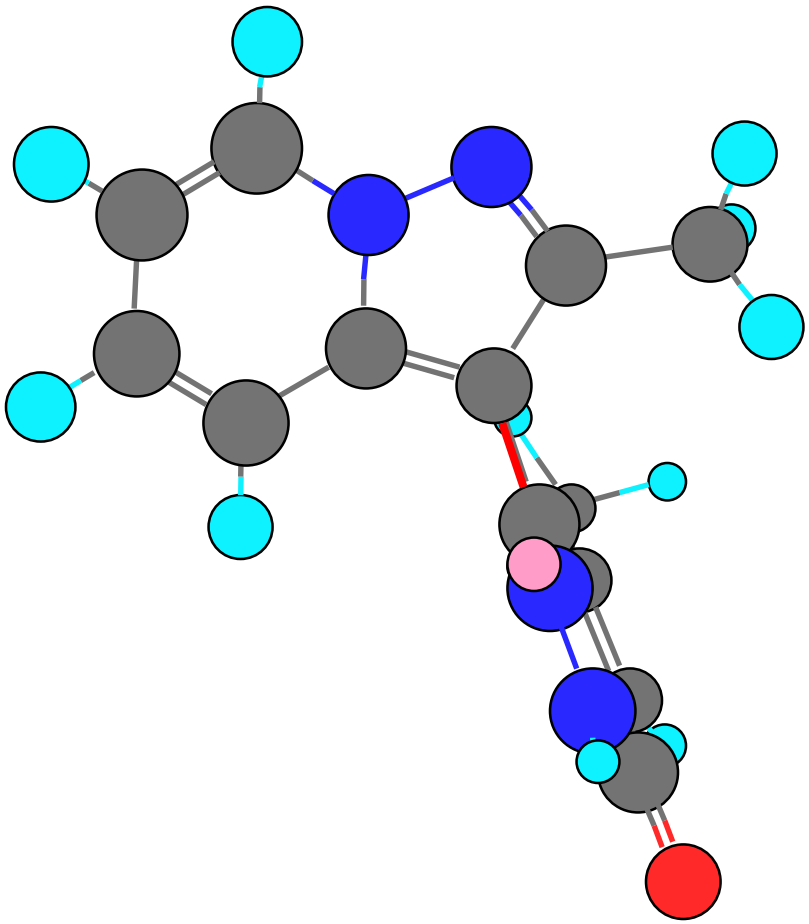
STERIC HINDRANCE

Redo 3d diagrams



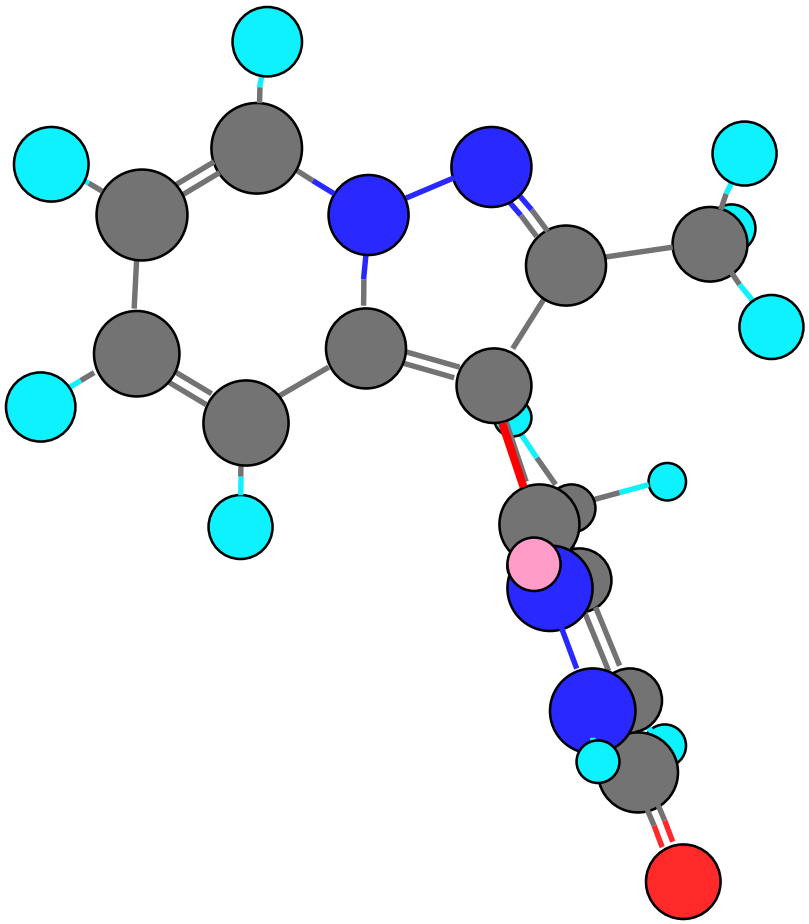
STERIC HINDRANCE

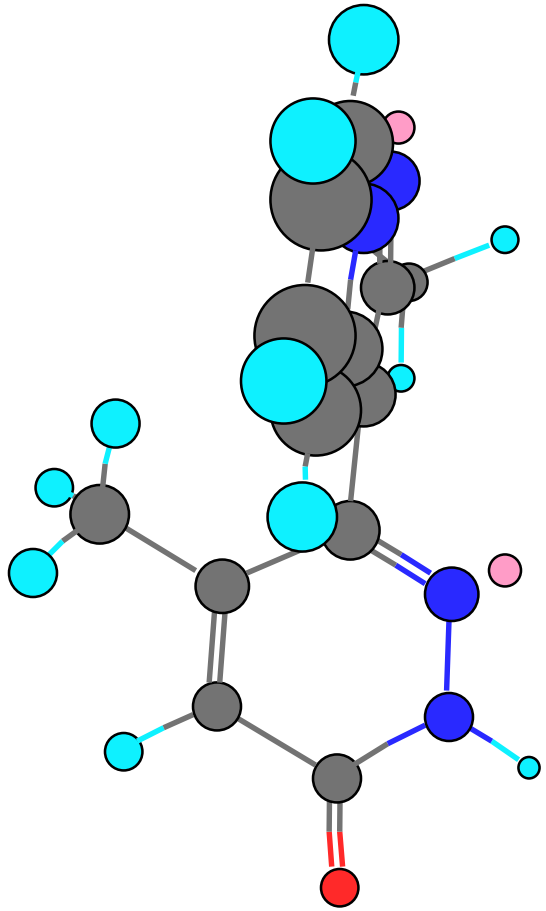
Redo 3d diagrams



STERIC HINDRANCE

Redo 3d diagrams





STERIC HINDRANCE

Redo 3d diagrams

Locked into orthogonal conformation

END OF LECTURE 1